THE RELATIONSHIP BETWEEN DATA VISUALIZATION

AND TASK PERFORMANCE

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Dissertation Prepared for the Degree of

DOCTOR OF PHILOSOPHY

UNIVERSITY OF NORTH TEXAS

December 2014

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Phillips, Brandon. *The Relationship between Data Visualization and Task Performance*. Doctor of Philosophy (Business Computer Information Systems), December 2014, 97 pp., 21 tables, 20 figures, references, 90 titles.

We are entering an era of business intelligence and big data where simple tables and other traditional means of data display cannot deal with the vast amounts of data required to meet the decision-making needs of businesses and their clients. Graphical figures constructed with modern visualization software can convey more information than a table because there is a limit to the table size that is visually usable. Contemporary decision performance is influenced by the task domain, the user experience, and the visualizations themselves. Utilizing data visualization in task performance to aid in decision making is a complex process. We develop and test a decision-making framework to examine task performance in a visual and non-visual aided decision-making by using three experiments to test this framework. Studies 1 and 2 investigate DV formats and how complexity and design affects the proposed visual decision making framework. The studies also examine how DV formats affect task performance, as measured by accuracy and timeliness, and format preference. Additionally, these studies examine how DV formats influence the constructs in the proposed decision making framework which include information usefulness, decision confidence, cognitive load, visual aesthetics, information seeking intention, and emotion. Preliminary findings indicate that graphical DV allows individuals to respond faster and more accurately, resulting in improved task fit and performance. Anticipated implications of this research are as follows. Visualizations are independent of the size of the data set but can be increasingly complex as the data complexity increases. Furthermore, well designed visualizations let you see through the complexity and simultaneously mine the complexity with drill down technologies such as OLAP.

ACKNOWLEDGEMENTS

First and foremost I want to thank my wife Angela, without whose untiring support, encouragement and care I could not have started and completed my Ph.D program.

I especially want to thank my dissertation chair Dr. Daniel Peak for his guidance, patience, support and encouragement throughout the program. I also thank my committee members Dr. Victor Prybutok and Dr. Charles Blankson for their insightful guidance to improve the content and presentation of my dissertation. I also thank my Ph.D coordinator Dr. Robert Pavur and my department chair Dr. Mary Jones for monitoring my progress at every stage of my program. I would like to thank all my professors, friends, and others who have directly and indirectly helped me complete this dissertation. I also thank University of North Texas for providing financial and academic supports that helped me achieve my goal.

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CHAPTER 1

INTRODUCTION

Early information systems (IS) research into data visualization (DV) primarily examined functional computer processes supported by basic interfaces, whose technology was capable of displaying only primitive graphics and unembellished text (G. W. Dickson, 1977). This study is motivated by the need to better identify the power of DV in the business computing environment (Peak, Prybutok, Gibson, & Xu, 2012). Most researchers recognize that astute professional and recreational technology users now expect rich sensory computer experiences, with DV playing an essential role (Wickham, 2013). Buoyed by the emergence of data analytics (Ware, 2012), sophisticated DV is now a staple of business intelligence (BI), where decision performance is driven by the task domain, the user experience, and the adeptness of the visualizations themselves (Gonzalez, 1996; Kobsa, 2004; Ware, 2012).

DV, as featured in BI software, enables users to analyze, display, and absorb large amounts of data in an effort to improve their decision outcomes (W. P. Dickson, 2005; Fayyad, Piatetsky-Shapiro, & Smyth, 1996; Ware, 2012). A graphical figure constructed with modern visualization software such as JMP can convey more information than a table because there is a limit to the table size that is visually usable and as a result it takes technical skill to reduce a large table to a manageable size when dealing with mega data. Whereas, a graphical representation can convey the same information in a more efficient, effective, and concise way (Tukey, 1990).

Drawing from Tufte et al. (1982; 1983), this research classifies DV to two broad categories: 1) functional DV, a spectrum where data is expressed as character-oriented (symbols, text, and numbers) tables, charts, and graphs and 2) aesthetic DV, where data is expressed as with aesthetic properties in an unlimited variety of interrelated diagrams, dynamic illustrations, and multi-dimensional graphic abstractions. Technology growth has extended functional charts and graphs so that they can now possess aesthetic characteristics (e.g., line, shape, color, perspective). As a result of the technology changes, the functional DV continuum extends into the aesthetic DV domain, which has the potential to benefit the decision maker. Although a decision maker may dissipate substantial cognitive effort in assimilating massive data that is depicted via the low-end, textual DV, in contrast the high-end, partially-aesthetic, graphical DV may require only a modest cognitive effort to clarify relationships, communicate patterns, and indicate trends not otherwise discernible (Daniel, 1992; W. P. Dickson, 2005; Feyyad, 1996; Robey & Farrow, 1982; Tufte & Graves-Morris, 1983).

The task performance model builds on key research that shows users pre-cognitively and instantaneously perceive visual information, thus adding to the traditional focus on how users cognitively process visual information (Lindgaard, Fernandes, Dudek, & Brown, 2006; Tufte & Graves-Morris, 1983). Hence, the task performance model describes for BI users a mental-energy-saving mechanism that simultaneously accelerates data assimilation and reduces cognitive load—without loss of decision or performance accuracy. As it examines the influence of two forms of DV on task performance, this research also explores the importance of individual factors to the BI decision process.

By integrating DV software into BI, contemporary BI systems become capable of providing ever-more powerful, varied, visually stimulating, and fruitful imagery to BI users (Ware, 2012). Because DV can have positive effects on decision performance and task performance, and because users differ in their individual characteristics, the present model integrates relevant decision theories with visual design, cognitive fit, and task performance theory to frame its research question:

RQ1: In performing perceptual tasks depicted with equivalent tabular DV and graphical DV methods, how does DV affect decision-making performance and task performance, when considering individual differences, task confidence, usability, emotion, and cognitive load?

CHAPTER 2

THEORETICAL FOUNDATION

This section describes the theoretical foundation of this research. The present task performance research model is based on fundamental decision theory that includes: 1) technology acceptance model (TAM) (Davis, 1989), which describes user acceptance of data visualization technology, 2) expectation confirmation theory (ECT) (Oliver, 1980), which describes continued usage of DV, and 3) information systems success (ISS) model (Delone, 2003; DeLone & McLean, 1992), which describes the success of decisions made using DV. The present model integrates these decision theories with visual design, cognitive fit, and task performance theory to support its premise: user acceptance and sustained engagement with the DV will promote task performance, and therefore the Business Intelligence system's success. Combining these theories can provide insight into the multifaceted relationships, which drive the success of a BI information system and yield understanding of individual and organizational impact on decision-making performance. We begin with a discussion of data visualization.

Data Visualization

Data visualization (DV) is a process that visually represents data by using applied visual design theory, techniques, and tools to reveal meaningful trends and patterns in the data (Cawthon & Moere, 2007; Cyr, Head, Larios, & Pan, 2009). By integrating DV software into BI, contemporary BI systems are capable of providing ever-more powerful, varied, visually stimulating, and fruitful imagery to BI users (Ware, 2012). As an essential BI feature, DV propels enriched, data driven decision making in the enterprise, thus compelling BI researchers to understand precisely what methods make visualization effective and what visual formats ultimately improve decision performance (Hassenzahl & Tractinsky, 2006; Zhu, 2007). Building upon the work of Cyr (2010; 2009; 2008), this research fills a gap in the decision-making literature involving task performance, centering on relationships between individual differences and presentation style (Cyr, Head, & Larios, 2010; Cyr, Kindra, & Dash, 2008). Specifically, this study adds to the DV research stream by developing a new task performance model and analyzing computer-aided DV decisions using that model, thus comparing individual task performance of subjects presented with both tabular functional DV and graphical functional DV having aesthetic characteristics (e.g, line, shape, color, perspective).

Because pre-millennial visual technology labored to display printed illustrations on a computer screen, much incipient DV research is centered on the technical aspects of the visualization algorithms themselves (Buja, Cook, & Swayne, 1996; Fayyad et al., 1996; Rivest, Bedard, & Marchand, 2001; Vesanto, 1999). Although inscribed illustration long preceded computer display technology, combining the established visual design knowledge with current dominant display technology can increase the flexibility of the graphic format and amplify the inherent communicative power of illustrative graphics (Peak, Gibson, & Prybutok, 2011). Graphical images are themselves instruments for reasoning about quantitative information, but of all the available methods for analyzing and communicating statistical information, well-designed data graphics are often the simplest and the most powerful (Tufte & Graves-Morris, 1983). Thus, a marvelous benefit of DV is the vast amount of quantitative information that can be funneled to users for efficient decisions, when it is presented graphically (Ware, 2012).

Graphical DV exploits natural human visual acuity, allowing users to grasp information quickly and accurately (Lindgaard et al., 2006; Lurie & Mason, 2007). In DV tasks, for users to evaluate the functionality of the DV requires several seconds, while perception of DV appearance is virtually instantaneous, so that users effortlessly form impressions and quickly bond with the design (Peter H Bloch, 1995; Daniel, 1992; Kelton, Pennington, & Tuttle, 2010). Ostensibly anticipating benefits of DV, researchers are studying less the functional and utilitarian aspects of decision making and have focused more recently on user visual effects and experiences (Cyr, Head, & Larios, 2010; Hassenzahl & Tractinsky, 2006). Another motivation for graphical DV research is the discovery that pre-cognitive, visual perceptions have long-lasting effects (Peter H. Bloch, 1995). In addition, the recent introduction of frameworks for examining aesthetics in the IT domain indicates this research is timely (Hassenzahl & Tractinsky, 2006). This paper draws from that bridging perspective by comparatively measuring user perceptions of equivalent information presented in both tabular DV and graphical DV formats. The study demonstrates that visual presentation plays a critical role in both decision-making and task performance by examining decision scenarios delivered with tabular DV and graphical DV stimuli.

Textual Versus Graphical Visual Concepts

User-processed functional DV, expressed as textual information in tables, charts, and graphs, requires varying degrees of cognitive effort to understand (Tufte & Graves-Morris, 1983). This research uses the term “textual” and “text” as synonymous with symbols, text, and numbers. Because of the disparate technological characteristics of early visual displays, the cognitive burden of analyzing displayed text was seen a multifactor problem (Dillon, 1992). While users still find the centuries-old practice of reading printed text the most accessible, they find reading displayed text more variable and difficult to process than printed text, such that displays can induce eyestrain and symptoms of illness (Bak & Meyer, 2011; Boschman & Roufs, 1997; Meyer, 2010). Physical factors such as text display brightness, clarity, and spatial arrangement of character information—to name a few—can have large effects on problem solving (Jang, Bell Trickett, Schunn, & Gregory Trafton, 2012).

This research compares experimental outcomes observed along the spectrum of functional DV, from tables to aesthetically-oriented graphs, because it recognizes that displayed text (including tabular text) can be problematic in its effects, while imagery is not subject to the same aforementioned limitations and has the ability to convey more information than textual displays. The authors investigate whether graphical imagery can convey visual information not readily-articulated textually, such that the results can be more precise and revealing than conventional statistical computation charts and tables (Tufte & Graves-Morris, 1983).

Cognitive Fit

Cognitive fit theory (CFT) provides a theoretical basis for understanding how information formatting supports decision-making tasks (Vessey, 1991). A cognitive fit, or mental match, occurs when the information accentuated by one particular presentation format (symbolic or spatial) matches the information required for a user to most easily complete the task (also, symbolic or spatial). This research subsumes symbolic and spatial formats under functional DV. Tables (i.e., symbolic formats) are most appropriate for presenting discrete sets of symbols (e.g., a table of transportation departure and arrival times), while graphs (i.e., spatial formats) are most appropriate for depicting relationships among discrete sets of symbols (e.g., change in GDP over time, or the relationship among the performances of a number of sales regions).

The cognitive fit focus has been most evident in graphs versus tables research (Benbasat & Dexter, 1985; Dickson, DeSanctis, and McBride, 1986), which indicates the influence of data representation on performance depends on "fit" with the task, thus supporting cognitive fit and task performance (Jarvenpaa, 1989; I. Vessey & Galletta, 1991). Speier and Morris (2003) examined query interface design and found that for simple tasks, text representation is preferred, while in complex tasks, visual representations are preferred. Furthermore, Lurie and Mason (2007) concluded that data visualizations can improve efficiencies in routine tasks and enable users to uncover new insights that would have otherwise been unnoticed. This research ascribes cognitive fit as having more relevance to non-aesthetic functional DVs, since cognition plays an important role in function-oriented decisions (Vessey, 1991), whereas it proposes cognitive fit as having less relevance to graphical DVs with aesthetic overtones because perception of the latter is precognitive.

Task Performance

In this research, task performance is a vital outcome variable that assesses how well the task was performed, conditioned by the overall effectiveness of the DV (Heath & Etheridge, 1991; Shneiderman, 1996; Tory, Kirkpatrick, Atkins, & Moller, 2006). Analogous to early DV research, most task performance research that dealt with visualization either centered on the technical aspects of producing the visualizations (Heath & Etheridge, 1991; Pal Singh, Gupta, & Levoy, 1994) or on assessing the more functional visualizations that predominated in the 1970s and 1980s (Ghai, 1981; Lucas, 1981; Tufte & Graves-Morris, 1983). Our study contributes to DV research in the post-millennial context by employing a contemporary, emerging DV tool (JMP by IBM SAS) to create our survey images.

This approach is supported by research indicating that DV components of BI increase usability and decision making effectiveness, which in turn increase the rapidity and accuracy of task performance (Cawthon & Moere, 2007). This holistic view of the task performance and decision-making has yet to be fully explored. One reason is that the technology of prior research is dated and constrained by the technology of the time, including limitations in display, software, and processing capability (Benbasat & Dexter, 1985; DeSanctis & Gallupe, 1984; Jarvenpaa, 1989). This research updates previous task performance and DV decision making research (Schmell & Umanath, 1988; Washburne, 1927; Watson & Driver, 1983) with current technology, filling a gap in the literature with fresh perspectives for a new generation of users. The holistic model integrates individual differences and perceptual characteristics to examine the decision process at the individual level.

Task Technology Fit (TTF)

The Delone and Mclean (1992) IS success model explains how the attributes of an IS system affect user satisfaction and ultimately use of the system. This work is basic to task technology fit (TTF), which examines success of the decision making process by proposing that task characteristics, along with the technology characteristics used in the system, join to create a fit (Goodhue & Thompson, 1995). The fit focus has been most evident in table versus graph research conditioned by individual differences, decision-making effectiveness, and task performance, where research indicates the data representation technique on task performance depends on the TTF with the task itself (Benbasat & Dexter, 1985; Dickson et al., 1986). TTF can account for enhanced task performance and improve system effectiveness by enabling a fit between the technology and the task itself (Zigurs & Buckland, 1998). Different types of visualization (e.g., 2d, 3d, mixed) can be effective for different types of tasks indicating that there is a “fit” between the task and the visualization that leads to optimal task outcomes (Tory et al., 2006).

To explore the determinants of task efficiency and effectiveness, IS research extends TTF to incorporate TAM constructs to explore system interface design (Mathieson & Keil, 1998). While TAM itself provides useful insight into user acceptance of an information system, it is not task focused. Correcting this shortcoming, Dipshaw and Strong (1999) created a task-focused model that integrated both TAM and TTF. The model's purpose was to investigate task performance of individuals who used software tools; results indicated that the integrated task-focused model exhibited more explanatory power, capturing ease of use and usefulness constructs from TAM and integrating with TTF to allow for a specific task-centered approach.

Individual Differences

Individual differences is an important component of task performance research because it examines the importance of in task perceptions, individual decision-making, and task performance. Three types of individual differences exist in the context of decisions: cognitive, personality, and demographic factors, all which work together to shape the individuals perceptions of the decision making task (Motowildo, Borman, & Schmit, 1997; Zmud, 1979). Individual differences are important because decision makers bring personal characteristics, including experience and attitudes toward the task, to the decision process which results in an individual decision. Individual differences have been the subject of many IS studies about their role in the acceptance of new technologies (Agarwal & Prasad, 1999; Van der Heijden, 2004; Venkatesh & Davis, 2000) . As with previous IS and DSS a research into task performance, this work recognizes the importance of individual perceptions of the task, including usability, emotion, confidence, and cognitive load (Zmud, 1979).

Individual differences research has centered on the acceptance of new IT (Harrison & Rainer Jr, 1992; Majchrzak & Cotton, 1988; Zinkhan, Joachimsthaler, & Kinnear, 1987). In that vein, research examined TTF and individual differences related to the use of mobile commerce and found that experience, cognitive style, and computer self-efficacy are major factors that contribute to task performance (Lee, Cheng, & Cheng, 2007). Other research indicated that problem-solving strategies and performance depend on the level of user system expertise and experience (Z. Lee, Wagner, & Shin, 2008). Others proposed a technology-to-performance chain model that used TTF, individual differences, as well as precursors of utilization such as attitude toward using, habit, social norms, and beliefs to explain task performance impacts (Goodhue & Thompson, 1995). Investigating factors that influence MIS success, (Zmud, 1979) extensively explored individual differences as they relate to MIS success. He concluded that not only do individual differences play a role in MIS success, but also cognitive behavior, MIS design user attitude, and user involvement. Outside of the IS discipline we see evidence of individual differences and their effects on task performance in a study in psychology by (Motowildo et al., 1997), which proposes a theory that suggests personality and cognitive ability variables have a significant effect on task habits, skill, and knowledge which in turn influence task performance. In the spirit of research, our study incorporates TTF and individual differences components into our data visualization task performance model in order to distill what the best fit is, as well as the individual decision maker differences that attribute to more favorable task outcomes.

Information Presentation

Building upon Cyr's work and utilizing the theories of cognitive fit and TTF (Cyr et al., 2009), this research constructs and tests a new task performance model to investigate the role of the decision maker, in the context of the decision-making process, to improve and recommend the fit needed to ensure successful task performance. While earlier studies were limited by the technology of the time, this research extends those concepts and incorporates new elements while utilizing a modern data visualization tool to create the visualizations. In visual computer aided decision making, the decision maker visually perceives and cognitively processes the revealed outcomes to generate informed business decisions. The information systems that support DV concepts and methods contain business-oriented data visualization objects, such as dashboards, OLAP cubes, and balanced scorecards, that facilitate decision-making (H. J. Watson & Wixom, 2007). Even basic computers and hand-held devices are vastly more powerful and capable of displaying complex graphics and animations. Now more than ever, developers need to understand the value of data visualizations and the process of creating them—how to blend visual form and system functionality to provide fresh sensory insights into massive, complex data.

Cognitive fit and information presentation have been jointly studied, specifically regarding the effects of graphical and tabular data representations and their effects on decision-making performance (Benbasat & Dexter, 1985; Ghai, 1981; Kelton et al., 2010). Kolata (1982) states that graphic displays enable scientists to utilize the uniquely human ability to recognize meaningful patterns in the data, and to see patterns that might otherwise have been imperceptible. Vessey (1991) proposes cognitive fit as a response to the graph versus table research stream and proposes that the relationship between task and information presentation format lead to increased task performance for individual users as a result of this fit. In line with this research, this research utilizes components of cognitive fit to identify the preferred fit with the DV presented in the experiment and how it relates to task performance, as measured by the rapidity and task accuracy.

CHAPTER 3

METHODOLOGY

Undergraduate and graduate students at a large university in the southwestern United States completed a 125-question survey in two randomly assigned groups administered online through Qualtrics survey administration platform. Each group consisted of a survey containing a pretest to measure the individual decision maker characteristics or individual differences, followed by two DV task. Each DV task was then followed by six questions related to the task and the visual aid used to complete the task. The subject will be graded for accuracy and timeliness with regards to the task. Finally, after the respondents completed the task they will answer demographics questions as well as questions concerning their perceptions of the tasks they have just completed.

Each survey group responded to two tasks representing equivalent information, one graphical DV and one tabular DV. We utilized a 2 X 2 survey design where task 1 contained information about mobile operating systems market share, while task 2 contained information about Titanic shipwreck survivors. The first group received graphical DV task information followed by tabular DV task information, while the second group received tabular DV task information followed by graphical DV task information. (Appendix).

This research design enables exploration of differences across the groups, and mitigates effects of the information presentation styles on the respondents. Respondents were assigned randomly to 1 of the 2 survey groups. The order in which the users were shown the visual and functional aids was randomized to mitigate order bias and increase reliability and validity of the sample. The survey ended up reaping 823 responses, which we used for data analysis.

Model Development Summary

This paper is the first to build and test a theoretical model that encompasses multiple variables to address DV and their effects on individual task performance. This experimental model aims to extend this current DV research and examine task performance in a visual aided decision making environment. This is explained in more detail in the following sections.

Survey Development Summary

The authors developed the survey by adapting items from previously validated research constructs. They piloted the preliminary version using an expert panel of IS faculty and doctoral students to validation the instrument and to gauge average completion time. Resulting comments and suggestions by the expert panel helped refine the final version of the survey, which the authors then distributed to students (please see Appendix A for the construct table). The chosen DV tool was JMP by IBM SAS (SAS Publishing, 2012) to prepare and create the visuals for the survey, in recognition that JMP is a rapidly expanding tool with increasing industry adoption (Gartner Group 2013). Gartner places DV at the peak of its 2013 hype cycle and places JMP by IBM SAS at position number 4 in the top five most used DV software, with 8% reported use and 35$ million on estimated revenue (Gartner 2013). According to a recent study by Gartner, the Business intelligence market increased 16% to 12.2 billion, and is expected to be 13.8 billion by the end of 2013.

Study 1

Study 1 was motivated by the static graphical versus tabular research in IS. The instrument constructed for this study included a graphical task as well as a tabular task for each of the 2 randomly assigned groups. The tasks were also randomized inside the instrument to control for order bias.

Study 2

Study 2 is motivated by the research into interactive displays of data versus static displays of data. For this study we compare a group of respondents who were given a survey instrument with only a interactive graphical or tabular display in order to assess how the DV format affected their task performance and timeliness.

Study 3

Study 3 is a meta-analysis of the results of study 1 and 2 in order to examine how the level of data visualization formats and complexity influence our visual decision making framework, and ultimate the perceptions constructs of usefulness, confidence, cognitive load, visual aesthetics, and emotion, and ultimate their effects on task performance.

CHAPTER 4

STUDY 1 STATIC TEXT VS GRAPHIC

Early IS research into data visualization (DV) primarily examined functional computer processes supported by basic interfaces, whose technology was capable of displaying only primitive graphics and unembellished text (Dickson, 1977). This study is motivated by the need to better identify the power of DV in the business computing environment (Peak, Prybutok, Gibson, & Xu, 2012). Most researchers recognize that astute professional and recreational technology users now expect rich sensory computer experiences, with DV playing an essential role (Wickham, 2013). Buoyed by the emergence of data analytics (Ware, 2012), sophisticated DV is now a staple of business intelligence (BI), where decision performance is driven by the task domain, the user experience, and the adeptness of the visualizations themselves (Gonzalez, 1996; Kobsa, 2004; Ware, 2012).

DV, as featured in BI software, enables users to analyze, display, and absorb large amounts of data in an effort to improve their decision outcomes (Dickson, 2005; Fayyad, Piatetsky-Shapiro, & Smyth, 1996; Ware, 2012). A graphical figure constructed with modern visualization software such as JMP can convey more information than a table because there is a limit to the table size that is visually usable and as a result it takes technical skill to reduce a large table to a manageable size when dealing with mega data. Whereas, a graphical representation can convey the same information in a more efficient, effective, and concise way (Tukey, 1990).

Drawing from Tufte et al. (1982; 1983), this research classifies DV to two broad categories: 1) functional DV, a spectrum where data is expressed as character-oriented (symbols, text, and numbers) tables, charts, and graphs and 2) aesthetic DV, where data is expressed as with aesthetic properties in an unlimited variety of interrelated diagrams, dynamic illustrations, and multi-dimensional graphic abstractions. Technology growth has extended functional charts and graphs so that they can now possess aesthetic characteristics (e.g., line, shape, color, perspective). As a result of the technology changes, the functional DV continuum extends into the aesthetic DV domain, which has the potential to benefit the decision maker. Although a decision maker may dissipate substantial cognitive effort in assimilating massive data that is depicted via the low-end, textual DV, in contrast the high-end, partially-aesthetic, graphical DV may require only a modest cognitive effort to clarify relationships, communicate patterns, and indicate trends not otherwise discernible (Daniel, 1992; Dickson, 2005; Feyyad, 1996; Robey & Farrow, 1982; Tufte & Graves-Morris, 1983).

While a full comparison of functional DV with aesthetic DV is beyond the scope of this study, this research compares experimental outcomes along the spectrum of functional DV—from tables to aesthetically-oriented graphs—thus filling a gap in the IS literature by demonstrating and explaining how graphical DV, is advantageous for BI decisions and task performance. Our task performance model builds on key research that shows users pre-cognitively and instantaneously perceive visual information, thus adding to the traditional focus on how users cognitively process visual information (Lindgaard, Fernandes, Dudek, & Brown, 2006; Tufte & Graves-Morris, 1983). Hence, the model describes for BI users a mental-energy-saving mechanism that simultaneously accelerates data assimilation and reduces cognitive load—without loss of decision or performance accuracy. As it examines the influence of two forms of DV on task performance, this research also explores the importance of individual factors to the BI decision process.

The present task performance research model is based on fundamental decision theory that includes: 1) technology acceptance model (TAM) (Davis, 1989), which describes user acceptance of DV technology, 2) expectation confirmation theory (ECT) (Oliver, 1980), which describes continued usage of DV, and 3) information systems Ssccess (ISS) model (Delone, 2003; DeLone & McLean, 1992), which describes the success of decisions made using DV. The present model integrates these decision theories with visual design, cognitive fit, and task performance theory to support its premise: user acceptance and sustained engagement with the DV will promote task performance, and therefore the BI system's success. Combining these theories can provide insight into the multifaceted relationships, which drive the success of a BI information system and yield understanding of individual and organizational impact on decision-making performance. We begin with a discussion of data visualization.

Model Development and Hypotheses

This paper is the first to build and test a theoretical model that encompasses multiple variables to address DV and their effects on individual task performance. This experimental model aims to extend this current DV research and examine task performance in a visual aided decision making environment.

The Cognitive Load Construct

Cognitive load, a key component of cognitive fit theory, is defined and measured as mental effort on the part of the decision maker (Paas, Van Merriënboer, & Schmidt, 2004). Cognitive fit elucidates that there is an ideal fit between the task information presentation and the task requirements in order for the decision maker to succeed. If a fit does not exist, then the task becomes more difficult and the cognitive load becomes greater. Copious research indicates that choices of graphic versus tabular information presentation significantly influence decision outcomes (Dickson et al., 1986; Jarvenpaa, 1989; Tukey, 1990; Vessey, 1991; Washburne, 1927). For this study we adopt the premise from the aforementioned research that while functional tabular representations are more effective for other tasks, they may not be for others.

The cognitive load construct is vital to proving a major premise of this research: Decision makers dissipate cognitive effort assimilating data depicted via functional DV, while they dissipate little or no cognitive effort pre-cognitively processing aesthetic DV. Functional DV is logically processed, which creates a cognitive load (Daniel, 1992; Dickson, 2005; Feyyad, 1996; Robey & Farrow, 1982; Tufte & Graves-Morris, 1983). On the other hand, research suggests that visual imagery is not logically processed (Peak et al., forthcoming; Peak et al., 2011), but instantaneously creates an impression that does not increase cognitive load.

H6: (A-G): Individual differences will have an effect on cognitive load

H10: Cognitive load will have a negative effect on task correctness.

H15: Cognitive Load will have a negative effect on Post Task Confidence

H14: Information presentation in a visual format will have a negative effect on cognitive load

H11: Perceived usability of the information presentation will have a negative effect on cognitive load

The Visual Design and Emotion Constructs

Although marketing and web design have examined visual design and resulting emotional response to the design for decades (Cyr et al., 2010; Cyr et al., 2009; Cyr et al., 2008; Lindgaard et al., 2006; Salisbury, Pearson, Pearson, & Miller, 2001), slight research in IS investigates this area.

Tractinsky (2004) argues that aesthetic design plays a major role in private, social, and business situations, positing that aesthetic design plays a key role in IT. Cawthon & Moere (2007) assert that aesthetics is a mechanism to explain and predict IS success, usability, and trust and to aid in the decision making process. In this study evidence suggests that across multiple visualization types, information presentations that were measured as being more visually appealing, scored higher on usability, and produced more correct responses. Much of the research into aesthetic design centers itself on website design and the user's perception of the site due to its perceived aesthetic value. These studies have shown that users respond positively to a website or interface perceived as aesthetically pleasing (Cyr et al., 2010; Cyr et al., 2008; Lindgaard et al., 2006). This research builds upon this stream by incorporating the visual design construct into our model framework; aesthetical perception of the DV design has been shown to influence the users' attitude, feelings of trust, and emotion.

Emotional responses to visuals and interfaces are equally as important as aesthetic value in influencing user's attitudes and feelings toward a system or visual aid. Beaudry & Pinsonneault (2010) found that emotions impact appropriate IS usage in work settings (Lindgaard et al., 2006). Other studies have found that users reactions to visuals are not only immediate but also invoke users to feel emotions in the form of sensations, pleasure and arousal (Osborne, 1979) indicating that users reactions to the medium does influence their attitudes towards the system. This research builds upon previous emotion response research by incorporating the emotion response construct into our model framework.

H12: Perceived visual design will have a positive effect on perceived usability

H13: Perceived visual design will have a positive effect on emotion

H4: (A-H): Individual differences will have an effect on emotion

H8: Pre-task confidence will have a negative effect on emotion

H13: Perceived visual design will have a positive effect on emotion

The Usability Construct

Usability is a cornerstone of the technology acceptance model (TAM) by determining whether an IT system is successful by being both easy to use and useful (Fred D Davis, 1989). This research applies TAM to decision-making systems (including BI) and DV. These systems facilitate the decision making process must also be deemed both easy to use and useful in order to be effective, and enable positive task outcomes (Doll, Hendrickson, & Deng, 1998). This research builds upon previous usability research by incorporating the usability construct into our model framework.

H5 (A-H): Individual differences will have an effect on usability

H12: Perceived visual design will have a positive effect on perceived usability

H16: Perceived usability will have a negative effect on post task confidence

H1: Information presentation in a visual format will have a positive effect on perceived usability

Table 1

Construct Table

Construct	Questions	Adapted From
Pre Task Confidence	I am confident in my ability to complete this task	Aldag and Power (1986)
	I am confident I will be able to complete this task	
	I am confident i will successfully complete the task	
Task Understanding	I understand the tasks instructions	Robey, Daniel, and Dana Farrow.1982
	I am knowledgeable of these tasks and what they require	
	It is clear to me what the tasks are asking me to do	
Computer Efficacy	I am very sure of my abilities to use computers.	Murphy, Christine A., Delphine Coover, and Steven V. Owen. (1989)
	I am very confident in my abilities to make use of computers.	
	I consider myself to be a skilled computer user.	
Attiude to task	These tasks are valuable	DeLone, William H., and Ephraim R. McLean. (1992)
	These tasks are interesting	
	I want to complete these tasks	
	I am willing to complete these tasks	
Task willingness	These tasks are important to me	Robey, Daniel, and Dana Farrow. (1982)
	I want to successfully complete these tasks	
	I want to perform these tasks well	
	These tasks seem exciting to me.	
Easy to use	It is easy to use the visual to accomplish my task	Li, Yung-Ming, and Yung-Shao Yeh. 2010 & Belardo. Kanvan. and Wallace (1982)
	The visual is easy to use.	
	The visual is easy to understand	
	The visual was relevant to the task	
Usefulness	The visual helped me be more effective in my task	Li, Yung-Ming, and Yung-Shao Yeh. 2010
	The visual made the task easier to complete.	
	The use of the visual required fewer steps to accomplish that I thought,	
	The visual helped me be more productive in completing the task.	

Table 2

Construct Measures

Construct	Questions	Adapted From
Emotion	The visual makes me feel stimulated	Deng and Poole (2010)
	The Visual makes me feel excited	
	The Visual makes me feel frenzied	
	The Visual makes me feel aroused	
	The Visual makes me feel jittery	
	The Visual makes me feel wide-awake	
Design Aesthetics	The visual design is attractive.	Li, Yung-Ming, and Yumg-Shao Yeh, (2010)
	The visual looks professionally designed	
	The overall look and feel of the visual is visual	
	The graphics used in the visual are meaningful	
	Overall I find that the visual presented looks attractive	
	The layout of the visual is attractive.	
Cognitive Load	How Mentally Demanding was the task?	Hart, Sandra G., and Lowell E. Staveland (1988)
	How hurried or rushed was the pace of the task?	
	How successful were you in accomplishing what you were asked to do?	
	How hard did you have to work to accomplish the task?	
	How insecure, discouraged, irritated, stressed, and annoyed were you with the task?	
Post task confidence	I feel confident that I completed the task corre	Gueutal.Surprenant, and Bubeck (1984)
	I feel confident in my task performance	
	I feel confident i completed the task successfull	

The Individual Differences Constructs

Individual differences are classified in IS research as cognitive, personality, and demographic factors (Zmud, 1979). Cognitive factors refer to the decision maker's cognitive style and preference in information processing (Messick, 1984). This research also examines personality factors which describe to the decision maker's confidence in the task, attitudes, computer and software self-efficacy (Compeau & Higgins, 1995). Demographic variables such age, gender, training, education, and skills document the diversity of the population. The four individual difference constructs in this model are personality factors: computer self-efficacy, attitude toward task, understanding, and task willingness. This research builds upon previous individual differences research by incorporating the aforementioned individual differences constructs into our model framework.

H3 (A-H): Individual differences will have a positive on pre task confidence

H8: Pre Task confidence will have a negative effect on emotion

H9: Post Task confidence will have a positive effect on task correctness.

Decision Making Process Framework

This research incorporates TAM, cognitive fit, TTF and individual differences theories to develop a framework that explores relationships in the decision making process, which influences task performance in functional DV and aesthetic DV environments. In developing this framework we grouped constructs into four major conceptual categories: task attributes, task perceptions, individual differences and task outcome (Figure 1).

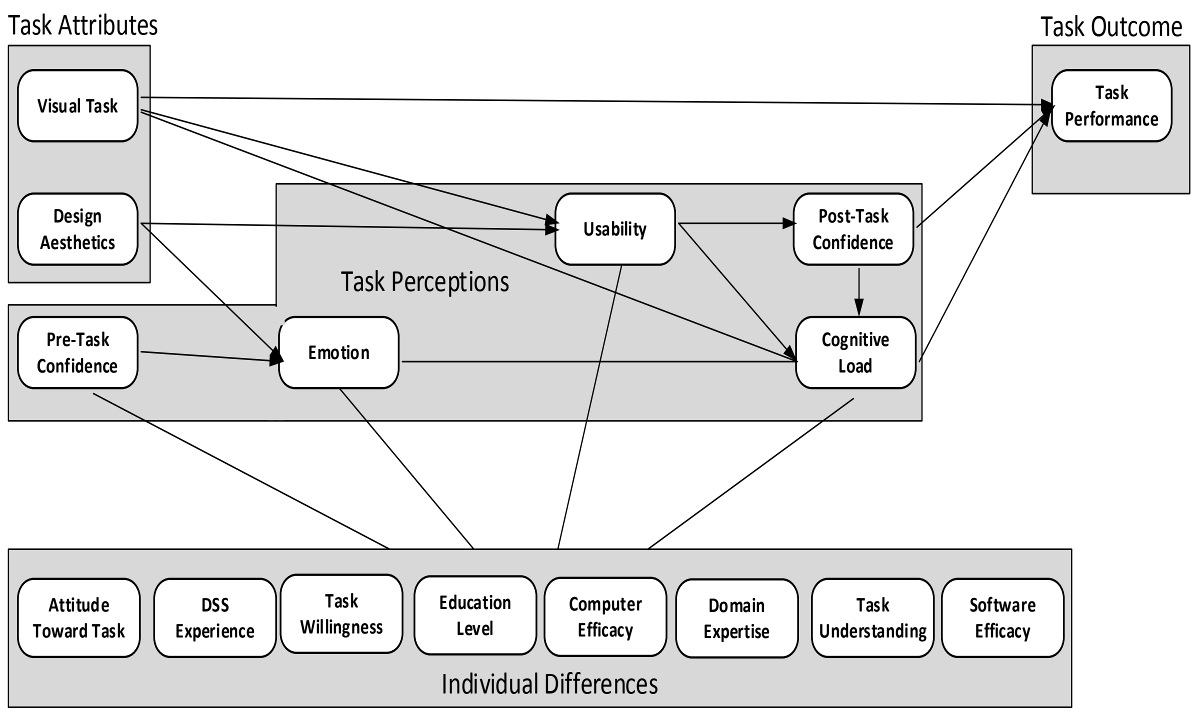


Figure 1. Decision making model framework.

Task attributes the first conceptual category, includes the constructs DV task (tabular or graphical) and visual design of the DV. Task perceptions, the second conceptual category, includes the constructs pre-task confidence, emotion, usability, cognitive load, and post-task confidence. Individual differences, the third category, includes the constructs attitude toward task, task willingness, computer efficacy, and task understanding. Task outcomes, the fourth and final category, includes the construct task performance Task outcome is our dependent variable and in this study includes task correctness, which is the percentage of correctly answered task questions. During the preliminary model development and pilot testing, results indicated task completion time to be insignificant, which is not surprising, as we designed the DV representations to be equivalent tabularly and graphically. Therefore, this research uses task accuracy as its sole measure of task performance. The initial testable decision making model can be seen in Figure 2

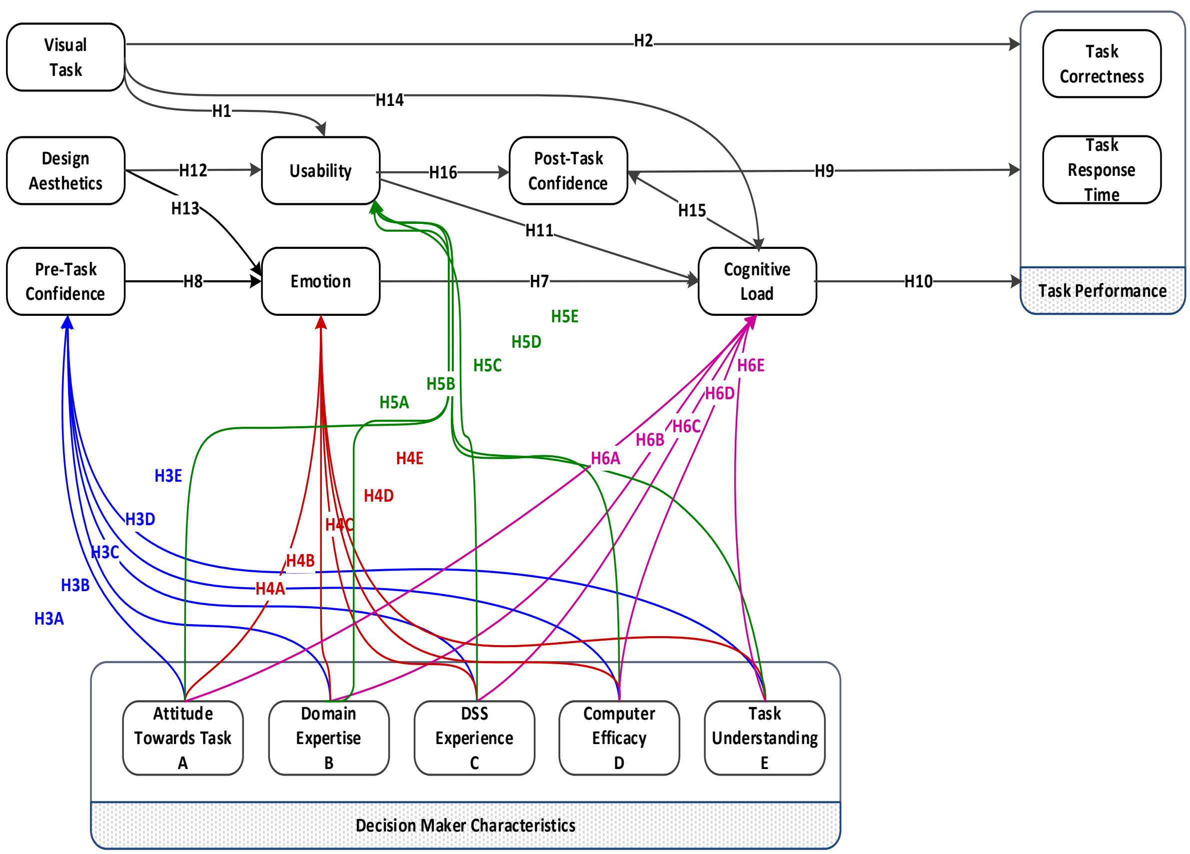


Figure 2. Initial testable decision making model.

Survey Development

The authors developed the survey by adapting items from previously validated research constructs. They piloted the preliminary version using an expert panel of IS faculty and doctoral students to validation the instrument and to gauge average completion time. Resulting comments and suggestions by the expert panel helped refine the final version of the survey, which the authors then distributed to students (please see table 2 for the construct table). The chosen DV tool was JMP by IBM SAS (SAS Publishing, 2012) to prepare and create the visuals for the survey, in recognition that JMP is a rapidly expanding tool with increasing industry adoption (Gartner Group 2013). Gartner places DV at the peak of its 2013 hype cycle and places JMP by IBM SAS at position number 4 in the top five most used DV software, with 8% reported use and 35$ million on estimated revenue (Gartner 2013). According to a recent study by Gartner, the Business intelligence market increased 16% to 12.2 billion, and is expected to be 13.8 billion by the end of 2013.

Methodology

Eight hundred twenty-three undergraduate and graduate students at a large university in the southwestern United States completed a 125-question survey in two randomly assigned groups administered online through Qualitrics survey administration platform. Each group consisted of a survey containing a pretest to measure the individual decision maker characteristics or individual differences, followed by a tabular and graphical DV task. The first task contained a graphical DV task and six questions related to the visual aid, which were graded for accuracy and timeliness. The second task was a tabular DV task where the respondents were again asked six questions and graded on accuracy and timeliness. Finally, after each task the respondents answered demographics questions as well as questions concerning their task perceptions.

Each survey group responded to two tasks representing equivalent information, one graphical DV and one tabular DV. We utilized a 2 X 2 survey design where task 1 contained information about mobile operating systems market share, while task 2 contained information about Titanic shipwreck survivors. The first group received graphical DV task information followed by tabular DV task information, while the second group received tabular DV task information followed by graphical DV task information. This is the same setup in each of the essays.

This research design enables exploration of differences across the groups, and mitigates effects of the information presentation styles on the respondents. Respondents were assigned randomly to 1 of the 2 survey groups. The order in which the users were shown the visual and functional aids was randomized to mitigate order bias and increase reliability and validity of the sample. The survey ended up reaping 823 responses, which we used for data analysis.

Preliminary Data Analysis

We performed a preliminary data analysis to assess the descriptive statistics in our study and to examine the demographic information of our respondents. Demographic information for all respondents is shown in Table 3. 58% of the respondents were male and 41% were female. 77.2% of the respondents in the study were between the ages of 18 and 25. Interestingly, 18% of the respondents indicated that they had some experience with decision support systems (DSS) and have at least had two statistics courses and either operation research or operations management course)

Table 3

Demographics

Gender			Age			Occupation			Student Classification			Major		
Male	478	58.1%	18-25	635	77.2%	Business	608	73.9%	Freshman	23	2.8%	Business	401	49.0%
Female	345	41.9%	26-34	140	17.0%	Science	12	1.5%	Sophmore	190	23.1%	Science	13	1.6%
			35-54	43	5.2%	Engineering	2	0.2%	Junior	377	45.8%	IT	84	6.8%
			55-64	4	0.5%	IT	78	9.5%	Senior	214	26.0%	Marketing	42	2.9%
			65+	1	0.1%	Other	60	7.3%	Graduate	19	2.3%	Logistics	59	3.2%
						None	63	7.7%				Accounting	69	3.6%
												Financing	92	4.8%
												Other	59	3.1%
Household Income			**Race**			**DSS Experience**			**Has Taken College Stats**			**Years of Work Experience**		
0-30K	352	42.8%	White	449	54.6%	Yes	148	18.0%	Yes	703	85.4%	None	104	9.5%
30-39K	72	8.7%	African Am.	113	13.7%	None	675	82.0%	None	120	14.6%	1-3 yrs.	310	28.2%
40-49K	65	7.9%	Hispanic	139	16.9%	**Stat Experience**			**Education Level**			4-6 yrs.	226	20.6%
50-59K	53	6.4%	Asian	73	8.9%	Yes	366	44.5%	Some Coll.	404	53.9%	7-9 yrs.	364	33.1%
60-69K	54	6.6%	Native Am.	7	0.9%	None	457	55.5%	2yr. Coll.	297	39.7%	10+ yrs.	95	8.6%
70-79K	34	4.1%	Pac.Island	5	0.6%	**BI Experience**			4yr. Coll.	48	6.4%			
80-89K	29	3.5%	Other	36	4.4%	Yes	194	23.6%						
90-99K	30	3.6%				None	629	76.4%						
100k>	134	16.3%												

Common Method Bias Analysis

Several techniques were used to address common method bias. First, standard statistical techniques such as Harman's one-factor test were used. Marker variables were also included along with the creation of psychological separation through the use of context shifts in the questionnaire (Craighead, Ketchen, Dunn, & Hult, 2011). Utilizing Harman's one-factor test we checked for the presence of common method bias (Harman, 1976). According to Podsakoff et al (2003), if common method bias is present, either a single factor will emerge from the factor analysis or one general factor will account for the majority of the covariance among the variables. The analysis did not indicate the emergence of a single factor nor did any one factor account for more than 50% of the covariance among the factors. Therefore, common method bias is not an issue.

Data Analysis and Results

We began our analysis by conducting an exploratory factor analysis (EFA) on the withheld half of the split sample data set. Results indicate that factor loadings are all above the recommended 0.7 (Henseler, Ringle, & Sinkovics, 2009) with all values greater than 0.83, indicating good discriminant validity (Henseler et al., 2009). A minimum value of 0.7 for Cronbach's alpha is desired (Nunnally & Bernstein, 1994) with values less than 0.6 indicating a lack of internal validity. For this study Cronbach's alphas are all greater than 0.7 indicating satisfactory internal reliability (Henseler et al., 2009).

Table 4

EFA Tabular Data

Scale Item	Std. Loading	Cronbach Alpha	AVE	Composite Reliability	Factor Correlations				
					1	2	3	4	5
TblEasyToUse1	0.92	0..942	0.852	0.958	**.849**	.121	-.049	-.149	.215
TblEasyToUse2	0.94				**.879**	.175	-.006	-.148	.135
TblEasyToUse3	0.93				**.882**	.156	-.014	-.117	.140
TblEasyToUse4	0.89				**.812**	.022	-.069	-.122	.256
TblUseful1	0.92	0.922	0.812	0.954	**.866**	.160	-.001	-.138	.197
TblUseful2	0.94				**.869**	.180	-.017	-.119	.146
TblUseful3	0.83				**.703**	.244	.098	-.114	.143
TblUseful4	0.92				**.814**	.225	.006	-.099	.229
TblEmotion3	0.90	0.822	0.822	0.948	-.037	.183	**.858**	.205	-.040
TblEmotion4	0.93				-.016	.253	**.871**	.186	-.069
TblEmotion5	0.93				-.065	.118	**.904**	.242	-.050
TblEmotion6	0.86				.076	.327	**.802**	.115	-.024
TblDesign1	0.92	0.812	0.813	0.955	.145	**.886**	.204	.095	.050
TblDesign2	0.91				.292	**.741**	.128	.033	.047
TblDesign3	0.91				.140	**.864**	.234	.072	.070
TblDesign5	0.93				.202	**.894**	.161	.059	.011
TblDesign6	0.93				.198	**.887**	.183	.094	.058
TblTaskCogLoad1	0.86	0.612	0.620	0.822	-.225	.176	.196	**.823**	-.047
TblTaskCogLoad2	0.82				-.070	.027	.161	**.842**	-.151
TblTaskCogLoad4	0.87				-.194	.107	.196	**.844**	-.059
TblTaskCogLoad5	0.86				-.227	.025	.178	**.837**	-.093
TblPostTaskConf1	0.96	0.894	0.895	0.962	.483	.075	-.085	-.143	**.820**
TblPostTaskConf2	0.96				.450	.079	-.062	-.153	**.839**
TblPostTaskConf3	0.91				.437	.073	-.067	-.125	**.848**

Tables 4, 5, and 6 show descriptive statistics, loading factors, and Cronbach's alphas for the tabular DV and graphical DV task as well as the individual differences. Results are all above the recommended thresholds for all constructs indicating reliability and validity of the data set. We move forward with the measures of the constructs that had the highest loadings and used those constructs to test our developmental decision making success model (Figure 2).

Table 5

EFA Visual Data

Scale Item	Std. Loading	Cronbach Alpha	AVE	Composite Reliability	Factor Correlations				
					1	2	3	4	5
VisEasyToUse1	0.99	0.951	0.873	0.965	**.856**	.173	.171	-.053	.264
VisEasyToUse2	0.99				**.869**	.194	.151	-.059	.230
VisEasyToUse3	0.99				**.880**	.171	.120	-.042	.213
VisEasyToUse4	0.99				**.764**	.225	.001	-.011	.230
VisUseful1	0.95	0.952	0.876	0.963	**.891**	.205	.071	-.031	.220
VisUseful2	0.94				**.894**	.189	.083	-.067	.192
VisUseful3	0.87				**.825**	.307	.153	.012	.089
VisUseful4	0.96				**.882**	.241	.114	-.042	.167
VisEmotion3	0.84	0.886	0.748	0.922	.073	.114	**.895**	.213	-.006
VisEmotion4	0.89				.215	.325	**.782**	.212	.086
VisEmotion5	0.91				.072	.173	**.861**	.335	.112
VisEmotion6	0.80				.291	.312	**.752**	.181	.086
VisDesign1	0.88	0.946	0.823	0.958	.216	**.863**	.112	.123	.011
VisDesign2	0.87				.282	**.828**	.168	.123	.103
VisDesign3	0.92				.238	**.890**	.163	.090	.056
VisDesign5	0.93				.253	**.868**	.215	.053	.108
VisDesign6	0.92				.261	**.844**	.220	.069	.070
VisTaskCogLoad1	0.82	0.870	0.718	0.910	-.044	.131	.223	**.839**	.051
VisTaskCogLoad2	0.79				.070	.081	.123	**.841**	-.106
VisTaskCogLoad4	0.81				-.091	.105	.261	**.826**	.063
VisTaskCogLoad5	0.85				-.193	.049	.165	**.783**	-.274
VisPostTaskConf1	0.97	0.967	0.939	0.979	.442	.079	.095	-.103	**.854**
VisPostTaskConf2	0.97				.444	.112	.085	-.096	**.846**
VisPostTaskConf3	0.97				.435	.106	.078	-.058	**.838**

Table 6

EFA Pre-Test Data

Scale Item	Std. Loading	Cronbach Alpha	AVE	Composite Reliability	Factor Correlations				
					1	2	3	4	5
PreConf1	0.94	0.942	0.896	0.963	.113	**.869**	.281	.218	-.047
PreConf2	0.95				.134	**.880**	.286	.189	-.063
PreConf3	0.93				.204	**.869**	.274	.119	.010
TaskKnowledge1	0.94	0.889	0.891	0.96	.180	.314	**.859**	.128	.013
TaskKnowledge2	0.94				.230	.264	**.864**	.155	.006
TaskKnowledge3	0.95				.198	.275	**.885**	.114	.014
CompEff1	0.95	0.927	0.871	0.953	.187	.238	.149	**.878**	.023
CompEff2	0.95				.186	.166	.150	**.911**	-.028
CompEff3	0.90				.206	.091	.073	**.877**	.074
TaskAttitude1	0.78	0.895	0.700	0.92	**.795**	.069	.197	.196	-.093
TaskAttitude2	0.85				**.845**	.084	.141	.169	.016
TaskAttitude3	0.80				**.802**	.099	.195	.112	.041
TaskInvolve1	0.87				**.827**	.131	.116	.114	.143
TaskInvolve4	0.87				**.821**	.144	.038	.115	.171

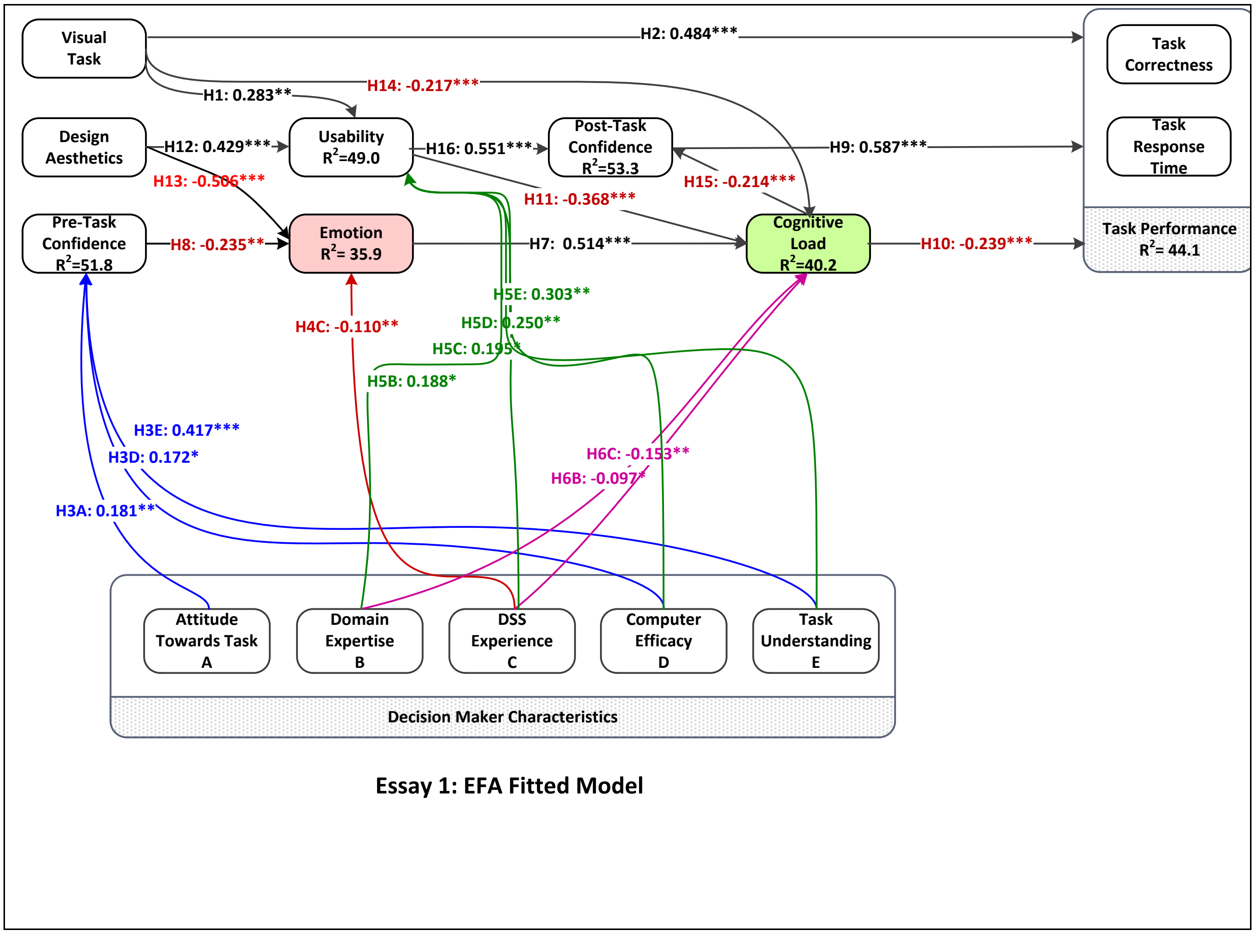


Figure 3. EFA developmental structural.

EFA Discussion

The EFA structural model derived from the withheld sample is shown in Figure 3 above. According to the data analysis, cognitive load, task confidence, and DV task account for 44.1% of the total variance associated with task performance. DV task, and visual design constructs together account for nearly 50% of the total variance associated with the usability of the visual aid. Individual decision maker characteristics, specifically, task attitudes, and computer efficacy are shown to have significant positive relationships with emotion and usability. The DV task, usability, and emotion constructs, together account for 40% of the total variance associated with cognitive load. Usability of the visual aid and cognitive load accounted for 53% of the total variance associated with post task confidence. Lastly, visual design, and task attitudes accounted for 36% of the total variance associated with the emotion responses elicited from the visual aid. The following figures represent the core framework model of decision making in the presence of data visualizations. Figure 4 represents the static tabular data visualization model and its effects on the core constructs in the framework. Figure 5 represents the static graphical data visualization model and its effects on the core constructs in the framework.

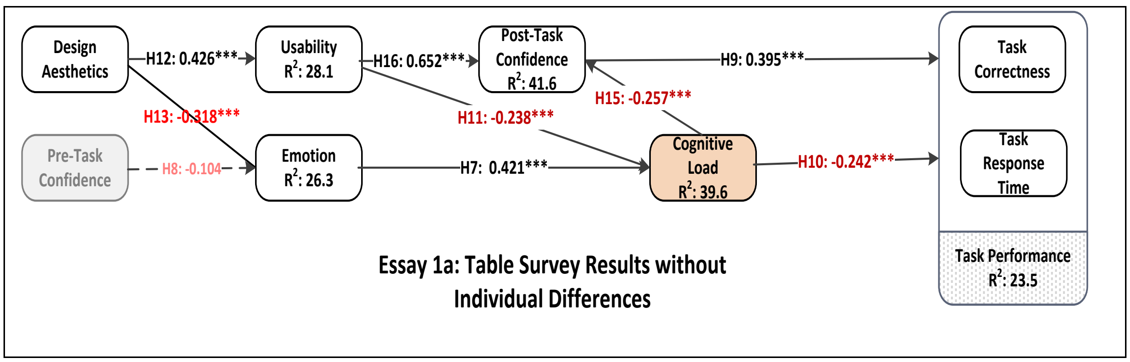


Figure 4. Developmental structural model with static tabular data visualization.

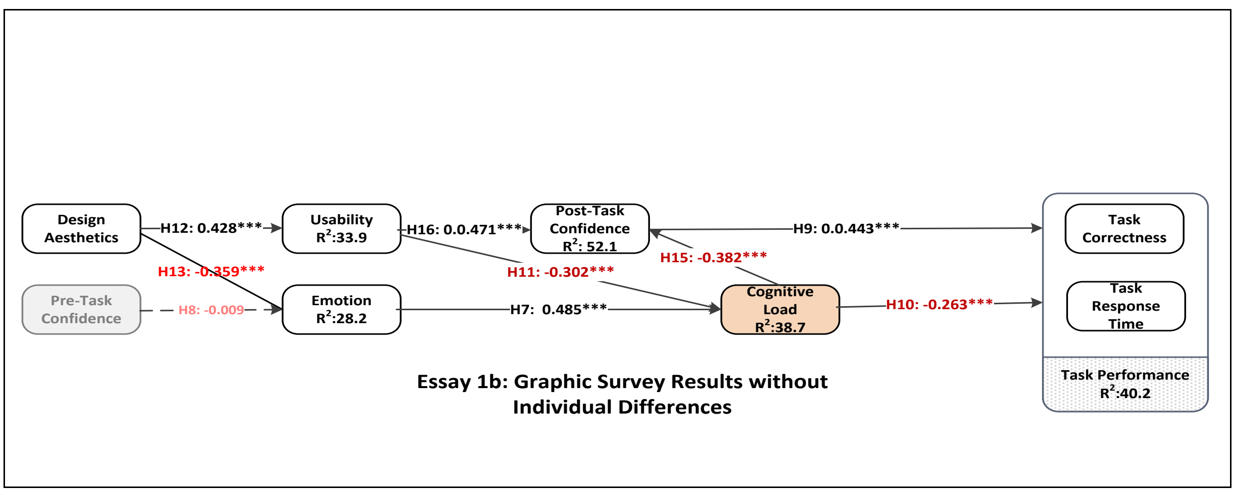


Figure 5. Developmental structural model with static graphical data visualization.

There are differences to be noted. In the tabular model, the r-squared of task performance is 23.5% while, in the graphical version, it is 40.2% indicating the using the graphical DV positively influenced task performance outcomes more than its tabular counterpart. Also, the core constructs (usability, emotion, post task confidence, and cognitive load) all exhibited a higher r-squared as well as path coefficients in the graphical model as opposed to the tabular model, indicating that the graphical DV model is a better fit.

Withheld Sample EFA Hypotheses Results

This EFA analysis supports strong relationship between individual differences on emotion (H7), and usability (H3). In addition, this EFA analysis finds that visual design positively influence information presentation usability (H12), and negatively effects emotion (H13). Further, graphical DV task also influences usability (H1) and overall task performance (H2). Results also indicate that cognitive load negatively influences task performance (H10), as well as post task confidence (H15). Interestingly, the relationship between the graphical DV task usability (H1), task performance (H2), and cognitive load (H14) exhibited a strong and significant relationship indicating that the graphical DV task allowed for higher task correctness, usability, and lower cognitive load.

Verification Sample CFA Data Analysis

For final model fit and verification, this study utilizes the split sample approach (Hair, 2001; Snee, 1977) to data analysis, where the second equal and random half of the whole data set is used to conduct a confirmatory factor analysis (CFA). Factor loadings are all above the recommended 0.7 threshold (Henseler, Ringle, & Sinkovics, 2009) with all values greater than 0.83 indicating good discriminate validity (Henseler et al., 2009). (Nunnally & Bernstein, 1991) recommend a Cronbach alpha minimum of 0.7with values less than 0.6 indicating a reduction in internal validity. In this study all Cronbach's alphas are greater than 0.7, indicating satisfactory internal reliability (Henseler et al., 2009). Tables 7, 8, and 9 show the descriptive statistics, loading factors, and Cronbach's alphas for the tabular DV and graphical DV task as well as individual differences. The CFA analysis confirmed the EFA analysis and chosen construct measures in both analyses were all above the recommended thresholds.

Table 7

CFA Tabular Data

Scale Item	Std. Loading	Cronbach Alpha	AVE	Composite Reliability	Factor Correlations				
					1	2	3	4	5
TblEasyToUse1	0.93	0.945	0.861	0.961	**.853**	.222	-.098	-.067	.162
TblEasyToUse2	0.95				**.878**	.192	-.100	-.054	.152
TblEasyToUse3	0.96				**.873**	.185	-.092	-.065	.180
TblEasyToUse4	0.87				**.805**	.057	-.017	-.082	.233
TblUseful1	0.94	0.944	0.856	0.959	**.885**	.136	-.163	.008	.138
TblUseful2	0.95				**.883**	.157	-.156	.029	.149
TblUseful3	0.88				**.799**	.172	-.184	.195	.021
TblUseful4	0.93				**.872**	.156	-.131	.065	.076
TblEmotion3	0.91	0.915	0.798	0.94	-.041	.157	.258	**.857**	.032
TblEmotion4	0.92				.000	.247	.185	**.862**	-.020
TblEmotion5	0.92				-.029	.122	.304	**.862**	-.056
TblEmotion6	0.84				.082	.352	.144	**.754**	-.001
TblDesign1	0.89	0.927	0.777	0.945	.165	**.839**	.130	.197	.037
TblDesign2	0.78				.214	**.738**	.044	.109	.114
TblDesign3	0.91				.178	**.856**	.065	.218	.025
TblDesign5	0.92				.207	**.886**	.071	.143	.032
TblDesign6	0.90				.167	**.867**	.100	.185	.031
TblTaskCogLoad1	0.91	0.909	0.785	0.936	-.190	.141	**.849**	.220	-.069
TblTaskCogLoad2	0.83				-.111	.072	**.821**	.224	-.001
TblTaskCogLoad4	0.91				-.225	.119	**.834**	.174	-.160
TblTaskCogLoad5	0.90				-.184	.077	**.793**	.273	-.192
TblPostTaskConf1	0.97	0.966	0.936	0.977	.597	.121	-.210	-.039	**.719**
TblPostTaskConf2	0.96				.579	.096	-.228	-.016	**.723**
TblPostTaskConf3	0.97				.583	.131	-.198	.004	**.735**

Table 8

Pre-Test Data

Scale Item	Std. Loading	Cronbach Alpha	AVE	Composite Reliability	Factor Correlations				
					1	2	3	4	5
PreConf1	0.93	0.927	0.873	0.953	.105	.295	**.858**	.183	.034
PreConf2	0.94				.175	.311	**.835**	.239	.032
PreConf3	0.93				.204	.234	**.867**	.171	.019
TaskKnowledge1	0.94	0.966	0.882	0.957	.140	**.858**	.297	.175	.050
TaskKnowledge2	0.93				.118	**.890**	.223	.134	.033
TaskKnowledge3	0.95				.162	**.884**	.270	.141	-.004
CompEff1	0.95	0.927	0.860	0.948	.228	.174	.202	**.878**	.010
CompEff2	0.94				.187	.184	.231	**.859**	.002
CompEff3	0.90				.202	.094	.125	**.876**	.130
TaskAttitude1	0.82	0.882	0.685	0.896	**.739**	.173	.196	.185	.033
TaskAttitude2	0.90				**.847**	.015	.064	.114	.059
TaskAttitude3	0.87				**.802**	.170	.196	.105	-.013
TaskWilling1	0.88				**.805**	.129	.085	.144	.159
TaskWilling4	0.91				**.786**	.036	.030	.181	.231

Table 9

Visual CFA Data

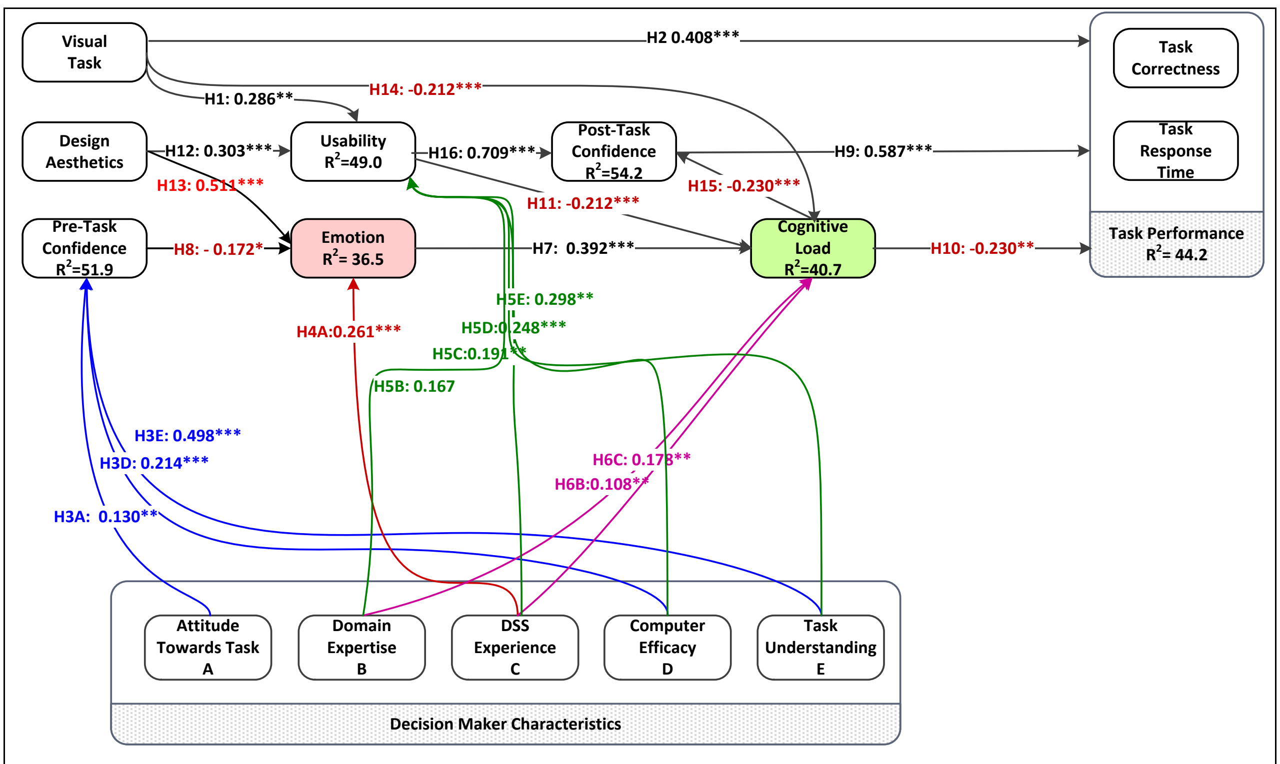
Scale Item	Std. Loading	Cronbach Alpha	AVE	Composite Reliability	Factor Correlations 1	2	3	4	5
VisEasyToUse1	0.99	0.990	0.98	0.996	**.867**	.160	.033	-.132	.265
VisEasyToUse2	0.98				**.796**	.315	.139	-.152	.144
VisEasyToUse3	0.99				**.833**	.245	.108	-.154	.208
VisEasyToUse4	0.99				**.872**	.163	.032	-.116	.215
VisUseful1	0.94	0.945	0.859	0.96	**.890**	.145	.019	-.091	.229
VisUseful2	0.95				**.900**	.176	.027	-.090	.203
VisUseful3	0.88				**.797**	.288	.171	-.112	.096
VisUseful4	0.94				**.863**	.213	.086	-.102	.189
VisEmotion3	0.79	0.885	0.75	0.921	.007	.141	**.787**	.272	.039
VisEmotion4	0.86				.157	.227	**.837**	.177	-.036
VisEmotion5	0.81				.035	.121	**.851**	.281	-.096
VisEmotion6	0.85				.236	.330	**.718**	.169	.017
VisDesign1	0.86	0.944	0.819	0.957	.177	**.848**	.164	.049	.111
VisDesign2	0.86				.221	**.811**	.174	.065	.045
VisDesign3	0.91				.253	**.869**	.165	.040	.030
VisDesign5	0.92				.245	**.878**	.150	.033	.121
VisDesign6	0.91				.277	**.860**	.155	.068	.080
VisTaskCogLoad1	0.81	0.873	0.73	0.912	-.149	.147	.192	**.861**	-.075
VisTaskCogLoad2	0.66				-.088	.057	.314	**.741**	.045
VisTaskCogLoad4	0.87				-.201	.071	.203	**.810**	-.226
VisTaskCogLoad5	0.86				-.215	-.053	.195	**.755**	-.245
VisPostTaskConf1	0.97	0.975	0.873	0.983	.441	.123	-.043	-.171	**.838**
VisPostTaskConf2	0.98				.463	.130	-.021	-.158	**.831**
VisPostTaskConf3	0.98				.434	.134	-.028	-.195	**.840**

Verification Sample CFA Discussion

We continue our analysis by using the second half of the randomized data sample to verify the model that was developed with the first half of the data to account for sample specific results. Further analysis and results through PLS as reference in Figure 6, indicate that developmental and verification model are similar in relationship significance and directionality. In the verification model task correctness has an r-squared of 58.5% which is derived from graphical DV task, post task confidence, and cognitive load. Together, graphical DV task, and visual design account for nearly 50% of the variance associated with usability. Cognitive load is influenced by usability, graphical DV task, and emotion which combine to account for 41.7% of the variance explained. The two individual differences, task attitudes and computer efficacy, have a similar effect on usability, and emotion from what was indicated in the developmental model.

Verification Sample CFA Hypotheses Results

This CFA also analysis confirms the strong relationship between individual differences on emotion (H7), and usability (H3). In addition, this study explores the role of visual design, and visual presentation with respect to usability of information presentations and cognitive load, and finds that visual design positively influence information presentation usability (H12), as well as emotion (H13). Further, graphical DV task representation also influence usability (H1) and overall task performance (H2). Results also indicate that cognitive load negatively influences task performance (H10), as well as post task confidence (H15). Interestingly, the relationship between graphical DV task representation and usability (H1), and task performance (H2), and cognitive load (H14) exhibited a strong and significant relationship indicating that the graphical DV task allowed for higher task correctness, usability, and lower cognitive load.



Essay 1: CFA Validated Model

Figure 6. Verification set structural model.

Construct Analysis

Table 6 shows the means for the constructs in the model related the tabular task versus the graphical task using the titanic data set. All construct mean differences are statistically significant. Compared to the tabular task, the graphical task measured better with regards to accuracy, usefulness, post task confidence, visual design, and emotional task engagement. The results showed an increase in task accuracy, and a reduction in task time and cognitive load, which suggests that respondents processed graphical DV more easily and that it provided a better task fit. Although all differences were statistically significant according to the levene's test, review of the mean charts indicate the largest magnitude of differences were task accuracy, time, and cognitive load with remaining differences being smaller. Table 6A shows the means for each of the major constructs.

Table 10

Construct Mean Table

	Static	Graphic
Decision Time	204	167
Decision Accuracy	77%	89%
Cognitive Load	3.1	2.6
Usefulness	4.2	4.7
Post Task Confidence	4.02	4.38
Visual Design	3.04	3.8
Emotion	3	2.8

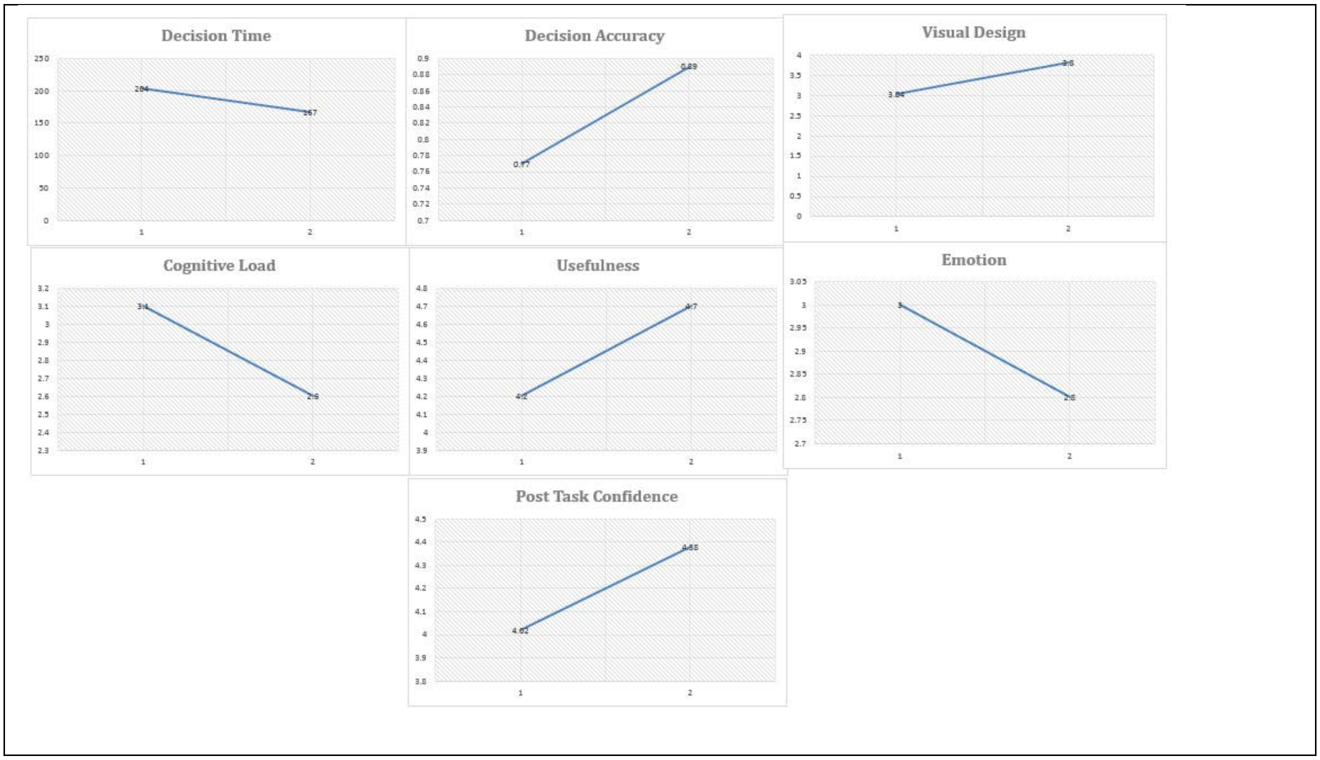


Figure 7. Respondent tabular (Column 1) versus graphical (Column 2) mean differences

Summary Discussion

The goal of this research is to develop a model of task performance and decision making that demonstrates the importance of visual graphics and individual factors and their influence on the decision making process through DV. In order to accomplish this, the study proposes and empirically tests a decision making success framework model based upon TAM, TTF, and cognitive fit in order to explain the dynamic and complex relationships involved in the decision making process and task outcomes.

The results indicate a strong relationship between individual differences and the emotion (H4), usability (H5), pre-task confidence (H3), and cognitive load (H6) constructs. Results also indicate that design graphics positively influences information presentation usability (H12), as well as emotion (H13), while graphical DV task positively influences usability (H1) and overall task performance (H2).

Table 11

Hypotheses Evaluation

	Hypothesis	Development Sample	Hold Out Sample
H1	Information presentation in a visual format will have a positive effect on perceived usability	Supported	Supported
H2	Information presentation in a visual format will have a positive effect on task correctness	Supported	Supported
H3A	Attitude towards task will have a positive effect on pre task confidence	Supported	Supported
H4A	Attitude towards task will have a positive effect on emotion	Not Supported	Not Supported
H5A	Attitude towards task will have a positive effect on usability	Not Supported	Not Supported
H6A	Attitude towards task will have a positive effect on cognitive load	Not Supported	Not Supported
H3B	Computer efficacy will have a positive effect on pre task confidence	Supported	Supported
H4B	Computer efficacy will have a positive effect on emotion	Not Supported	Not Supported
H5B	Computer efficacy will have a positive effect on usability	Supported	Supported
H6D	Computer efficacy will have a positive effect on cognitive load	Not Supported	Not Supported
H3C	Task understanding will have a positive effect on pre task confidence	Supported	Supported
H4C	Task understanding will have a positive effect on emotion	Not Supported	Not Supported
H5C	Task understanding will have a positive effect on usability	Supported	Supported
H6C	Task understanding will have a positive effect on cognitive load	Not Supported	Not Supported
H7	Emotion elicited by the information presentation will have a positive effect on cognitive load	Supported	Supported
H8	Pre Task confidence will have a negative effect on emotion	Supported	Supported
H9	Post Task confidence will have a positive effect on task correctness.	Supported	Supported
H10	Cognitive load will have a negative effect on task correctness.	Supported	Supported
H11	Perceived usability of the information presentation will have a negative effect on cognitive load	Supported	Supported
H12	Perceived visual design will have a positive effect on perceived usability	Supported	Supported
H13	Perceived visual design will have a negative effect on emotion	Supported	Supported
H14	Information presentation in a visual format will have a negative effect on cognitive load	Supported	Supported
H15	Cognitive Load will have a negative effect on Post Task Confidence	Supported	Supported
H16	Perceived usability will have a negative effect on post task confidence	Supported	Supported

Cognitive load negatively influences task performance (H10), as well as post task confidence (H15). The relationship of graphical DV task with usability (H1), task performance (H2), and cognitive load (H14) is strong and positively significant, indicating that the graphical DV task allowed for higher task correctness, usability, and lower cognitive load.Table 11 shows the comparative results of the withheld sample set (EFA) and verification sample set (CFA) hypotheses evaluation and their corresponding significance and support. Each hypothesis that was supported and significant in the withheld sample, also showed support and significance in the model verification sample, indicating that the resulting model and their results are not sample specific and giving further validation to our decision making success model.

The Importance of the Graphical DV Task

In both the developmental and verification models, the relationship between the graphical DV task and task correctness was highly significant and that relationship alone, accounted for more than half of the variance explained in task performance. We conducted an ANOVA using PLS regression where the graphical DV task was indicated by a 1 and the tabular task indicated by a 0.

Conclusion

This research fills an important gap regarding the effects of data visualization on task outcomes. Building on TAM, TTF, cognitive fit, and individual differences theories integrated with DV, this research achieves support for its premise: in a computer decision scenario, both user acceptance of and sustained engagement with the DV enhance task performance, and therefore promote the BI system's success. The results also confirm our tenet that DV shapes task outcomes by guiding task attitudes and task perceptions, which empower users to display, analyze, absorb, and act on large amounts of data and thus improve their task performance.

Finally, this research study confirms that graphical DV allows individuals to respond faster and more accurately and as a result improves task fit. Task fit, which ensures enhanced task performance (Dishaw & Strong, 1999; Goodhue & Thompson, 1995), is created in this study when the DV aligns with the attributes and perceptions of the decision maker. With these findings, this research extends the defined role of DV to embrace purely functional tables and graphical aesthetic design as critical features of the contemporary BI decision process.

As this study fills the DV research gap, it provides new knowledge by developing and testing a holistic theoretical model that combines visualization and perception with decision making and task performance of functional and aesthetic DV. This research extends the current literature by showing: 1) additional benefits of design-aesthetic stimuli over text-based stimuli to the overall task, 2) effects of graphical visual design on reducing cognitive load, and 3) the essential contribution of visual design to improving task performance and outcomes. This study further extends the body of knowledge by exploring the many dimensions of decision making and the decision makers' role and influence in the process.

Results of this study confirm that the in addition to individual differences, the graphical DV task was shown to be an important influencer of overall task performance as well as the task perceptions indicated by the user. The supported hypotheses verify the major premise of this study that not only do individual differences effect task performance, but so do information presentation, attitudes, and graphical visual design. This study emphasizes the importance of graphical DV tasks in a decision making model, more so than prior research has implied in this area. In addition to the importance of the graphical DV task itself, cognitive load, and perceived usability of the visualizations indicate a strong relationship concerning task outcomes, mainly task performance. This evidence indicates that a graphical DV task takes less mental effort (cognitive load), and is perceived more usable by the end user, which translate to improved task performance and a more accurate and efficient decision being made.

Limitations

Despite the fact that undergraduate students are a useful sample population due to their comfort and use level of technology as well as the fact that they will be future users of technology, the homogeneity of the sample is a limitation for a number of individual differences that were significant. A more diverse sample set would yield more individual differences than in this study. This would be beneficial to the model and the decision making process itself, as there would be more predictors of successful task outcomes, task attitudes, and task perceptions. A second limitation is the use of static two-dimensional data visualizations for the task. While this was effective for the purposes of our study in order to develop and test our model, the use of dynamic two dimensional, three dimensional DVs may yield more insight with future research.

Future Research

The results of this study lead us to suggest several areas for future research. One area would be to further test this model with a more diverse sample. The homogeneity of the student sample used in this study, made the individual differences not as significant as hypothesized. A more diverse sample may identify additional individual differences that influence task perceptions and decision making. Another area to explore is the use of different types of DV (interactive, dynamic, three-dimensional and two-dimensional) visual aids, in the decision making process, and how they differ from the graphical aesthetic designs used in the study. This study focused on the first step of building a DV developmental model. Subsequent studies should investigate the varying types of DV used in decision making in the enterprise.

CHAPTER 5

STUDY 2 INTERACTIVE TEXT VS GRAPHIC

Introduction

The decision making process has been at the forefront of research for decades. With the evolution of decision support system (DSS) from executive information systems (EIS), to management information systems (MIS) and beyond, the process by which decisions are made in organizations today using these systems is of utmost importance to the researcher and practitioner. As DSS are becoming ubiquitous and evolving into business intelligence systems which seek to empower decision makers to make important decisions, it is vital that the decision is making process, and with it, the decision maker be examined, in order to enable the highest level of task performance success and encourage the partnership of the decision maker with the decision task that they will most excel at.

Theoretical Foundation and Hypothesis Development

Research into DSS has its roots in the acceptance of the technology through Fred Davis's (1986) technology acceptance model (TAM), and the continued usage of the system by utilizing expectation confirmation theory (ECT) proposed by (Oliver, 1980). Research into decision making success is then explored more by the Delone and Mclean information systems success model in 1992 and again in a ten year update in 2003. This model proposes a means by which to understand the multifaceted relationship which drives the success of an information system, and to explain individual and organizational impact on decision making performance.

The Delone and Mclean model will serve as the groundwork for task technology fit (TTF) (Goodhue, 1995; Goodhue and Thompson, 1995) which goes further into examining the decision making process, by proposing that task characteristics, along with the technology characteristics used in the DSS, go together to create a fit that enhances task performance. TTF has been used in conjunction with constructs from TAM to explore DSS Interface design to enable users to perform tasks more efficiently and effectively (Mathieson and Keil, 1998). Another study done by Dishaw and Strong (1999) conceptualized a new integrated model derived from both TAM and TTF which sought to explain software utilization and its effect on user performance by indicating that the combined model explained more of the variance in software utilization by capturing ease of use and usefulness from TAM and integrating those constructs with TTF. Other studies have shown that TTF can be used to account for enhanced group task performance and improve DSS effectiveness (Zigurs et al, 1999; Zigurs and Buckland, 1998).

More in line with this paper's study is the research in the area of TTF and individual differences, (Lee et al, 2007) examined TTF and individual differences related to the use of mobile commerce and found that experience, cognitive style, and computer self-efficacy are major factors that contribute to task performance. Lee et al 2008 explores system experience and expertise further in his study and finds that problem solving strategies and performance depended significantly on the user's level of system expertise and experience. Another similar study conducted by (Goodhue and Thompson, 1995) proposes a technology to performance chain model which uses TTF, individual differences, as well as precursors of utilization such as attitude toward using, habit, social norms, and beliefs to explain task performance impacts. Many factors are believe to influence MIS success, Zmud 1979 extensively explored individual differences as they relate to MIS success and concluded that not only do individual differences play a role in MIS success, but so does cognitive behavior, MIS design user attitude, and user involvement. Outside of the IS discipline we see evidence of individual differences and their effects on task performance in a study in psychology by Motowidlo 1997, which proposes a theory that suggests personality and cognitive ability variables have a significant effect on task habits, skill, and knowledge which in turn influence task performance.

Cognitive fit and information presentation are other areas where a great deal of research is done with regards to DSS and the decision making process. A considerable amount of research has been done in the effects of graphical and tabular data representations and their effects on decision making performance. Kolata 1982 stated that graphic displays enabled scientist too make use of the uniquely human ability to recognize meaningful patterns in the data, and to see patterns that would of otherwise not have been seen. Vessey 1991 proposed cognitive fit as a response to the graph versus table research stream and proposed that the relationship between task and information presentation format leads to increased task performance for individual users. Information presentation has long been recognized as an important part of the decision making process (Kelton et al, 2010). In their model Kelton et all propose that decision maker characteristics as well as information presentation, mental representation and task characteristics are strong indicators of decision making outcomes Cognitive fit can be viewed as an evolution of Task technology Fit as it proposes that the task, whether graphical or tabular can display the same underlying information, however, due their nature, they are interpreted differently by the decision maker, and therefore the task solution is dependent upon a fit of the task presentation and the decision makers mental model. Vessey's research indicated that when the information presentation type matched the questions being asked, task performance increased. Also, participants in his study preferred tables rather than graphs and preferred symbolic rather than spatial problems. Drawing upon cognitive fit, and prior information presentation research Lurie and Mason (2007) propose a model to explain the implications of visual representation in the decision making process. Their model takes into account the visual perspective and information context of the visual representation. In addition, there model also uses user characteristics, or differences in order to explain task performance and outcomes. Following this research stream, other studies such as (Zhu 2007) have attempted to quantity what effective data visualization is and how one can be measured. He comes up with a measurement tool consisting of Accuracy, utility, and efficiency in order to properly measure a visualizations effectiveness value. When a visual is deemed valuable or effective, Li and Yeh 2010 concluded in their study that design aesthetics of a visual or system leads to increased trust in the system by the user.

By looking at the decision making research and how it has evolved, we begin to see a clearer picture of the multifaceted decision making process, and the variables that influence the outcome performance and user interactions. It is with this lens in mind that this paper proposes a decision making success model that incorporates TAM, TTF, as well as individual differences,

Information Presentation

Information presentation has been researched extensively as it relates to how decision makers interpret and process information delivered in a variety of formats, mainly graphic and tabular. Early research has shown that graphic displays enable decision makers to see patterns in data that would have otherwise been very difficult to see in a textual format (Kolata 1982 p. 919). Researcher have used a variety of terms from “information visualization” (Card, Mackinlay and Shneiderman 1999), to “data visualizations” (Green 1998) to refer to information presented to the decision maker in a visual form. Vessey 1991, proposed cognitive fit theory to explain why information presentation is vital to task performance. Cognitive fit tells us that the information presentation and task requirements must be matched in a way to create a fit, which enables positive task performance outcomes. Another study proposes that information presentation influences the perceived usefulness and ease and by creates a better task fit in order to increase performance. (Mathieson and Keil 1998).

Visualization is necessary to explore and analyze large amounts of data. The advantage of data visualization is that the user is directly involved in the mining process through interactive visualizations (Keim, 2002). Interactive visualization allow for easier navigation and exploration of data, as well as hypotheses testing, something that is more time consuming to do with static visualization and increasingly difficult to do with vast amounts of data (Grinstein & Ward, 2002). This study utilizes interactive data visualizations and explores their influence on task performance and the decision making process. A majority of the interactive data visualization research deals with creating the DV (Buja, McDonald, Michalak, & Stuetzle, 1991; Ward, Grinstein, & Keim, 2010). It is the aim of this research to investigate the effects of interactive visualizations on task performance in the context of an individual decision making process. Data analysis is fundamentally an iterative process in which you submit a query, receive a response, formulate the next query based on the response, and repeat. You usually don't issue a single, perfectly chosen query and get the information you want from a database. Therefore the purpose of data analysis is to extract unknown information, and in most situations, there is no one perfect query. People naturally start by asking broad, big-picture questions and then continually refine their questions based on feedback and domain knowledge (Hellerstein et al., 1999). This research explores the way in which users can utilize the interactive visualization to explore the data and answer specific questions regarding the dataset, that we hypothesize would be more difficult and time consuming given a static non interactive data visualization. As the extant research suggests, utilizing an interactive will allow for faster data discovery and querying that should enhance time to completion and accuracy.

Individual Differences

The influence of individual differences on information system use has been studied extensively. Zmud 1979 classifies individual differences as cognitive factors, demographics, and personality. Cognitive factors refer to the decision maker's cognitive style and preference in information processing (Messick 1984). Most common type of cognitive styles include intuitive, analytics, as well as learning styles such as visual, aural, read/write, and kinesthetic. Personality factor refers to the decision maker's confidence in the task, computer and software self-efficacy (Compeau and Higgins, 1995). Computer self-efficacy refers to the decision maker's level of confidence or belief in their software, or computer ability. Demographic differences that influence the use of information technology include variables such age, gender, training, education, and skills.

Cognitive Load

Cognitive load is defined and measured as mental effort on the part of the decision maker (Pass and Van Merrienboer, 1994). As cognitive fit explains that there is an ideal fit between the task information presentation and the task requirements in order for the decision maker to succeed. By extension, if this "fit" does not exist then the task becomes more challenging to complete successfully. Much research has been done in the area of graphic versus tabular information presentation and their effects on the decision outcome. The main theme being that graphic representations are a fit for some tasks, while tabular is more effective for other tasks. The decision maker's individual differences also play a role in their information acquisition of the information presented in these various formats (Vessey 1991).

Visual Design Aesthetics and Emotion

Visual design aesthetics and emotional response to the design has been something that has been researched and utilized in marketing and web design for decades. Until more recently, Aesthetics has been under represented in the MIS discipline. It has been found in the MIS discipline as a way to explain and predict MIS success, usability, and to aid in the decision making process. In a study by Cawthon and Moere (2007) they found that across multiple visualization types, the information presentations that were measured as being more aesthetically appealing, scored higher on usability, and produced more correct responses related to the task.

Usability

Usability has been around since TAM was conceptualized by Fred Davis in 1986 and proposes that in order for a system to be successful it must be easy to use and useful. In the decision making process realm, this is still very true, the technology and information presentations that facilitate the decision making process must be deemed useful and easy to use and understand in order for them to be effective, and enable positive task performance.

Task Attitude, Involvement, and Understanding

Decision maker's attitudes toward the system, and their subsequent involvement and willingness to participate and complete the task is another important part of the multifaceted decision making process. Zmud explored this in his 1979 study on individual differences and MIS success. User involvement and MIS success has been explored by Ives and Olson in a 1984 study they concluded that post implementation involvement and attitudes toward the DSS, were indicators of success for the system. Similar findings are discussed in Kaiser and Scrinivasan (1980), and Lucas (1975, 1976), as well as Maish (1979). They all concluded that user attitudes and beliefs toward the system indicated successful outcomes. With the literature as our guide and by incorporating TAM, TTF, cognitive fit, and Individual differences theories, we build our theoretical model and subsequent hypotheses in Figure 7.

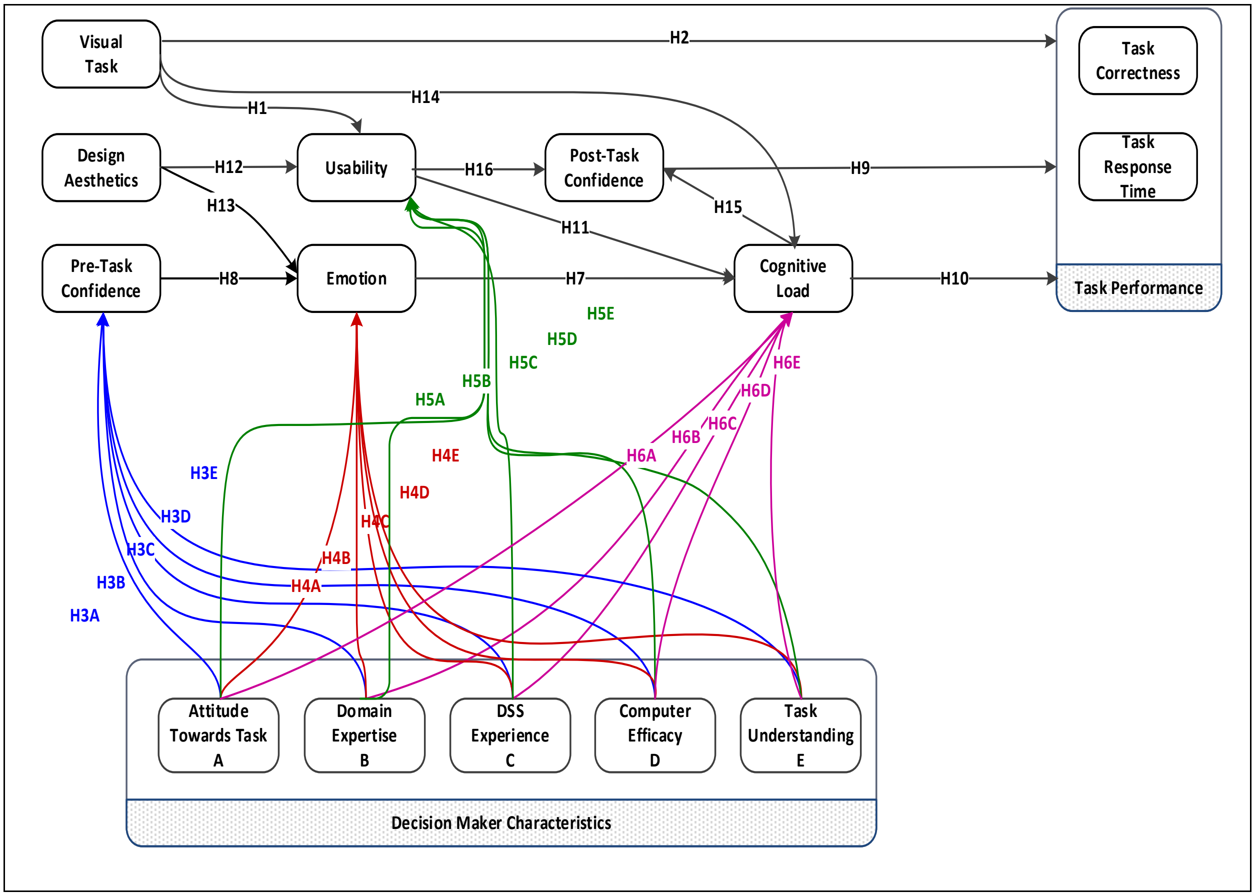


Figure 8. Testable decision performance model

Table 12

Hypotheses

	Hypothesis
H1	Information presentation in a visual format will have a positive effect on perceived usability
H2	Information presentation in a visual format will have a positive effect on task performance
H3(A-H)	Individual differences will have an effect on pre task confidence
H4(A-H)	Individual differences will have an effect on Emotion
H5(A-H)	Individual differences will have an effect on Usability
H6(A-H)	Individual differences will have an effect on Cognitive Load
H7	Emotion elicited by the information presentation will have a positive effect on cognitive load
H8	Pre Task confidence will have a negative effect on emotion
H9	Post Task confidence will have a positive effect on task performance
H10	Cognitive load will have a negative effect on task performance
H11	Perceived usability of the information presentation will have a negative effect on cognitive load
H12	Perceived design aesthetics will have a positive effect on perceived usability
H13	Perceived design aesthetics will have a negative effect on emotion
H14	Information presentation in a visual format will have a negative effect on cognitive load
H15	Cognitive Load will have a negative effect on Post Task Confidence
H16	Perceived usability will have a Positive effect on post task confidence

Methodology

Data was collected using a survey instrument administered online through qualitrics to undergraduate and graduate students at a large university in the southwestern United States. The survey consisted of 125 questions and two survey groups. Each survey group consisted of a survey containing a pretest to measure the individual decision maker characteristics or individual differences, followed by an interactive visual and interactive tabular information presentation task and demographic questions.

The first task contained an interactive visual information representation and 6 questions concerning the visual which were graded for accuracy and response time. The respondents were then asked a serious of questions about the task they completed. The second task was an interactive tabular information presentation task where the respondents were again asked 6 questions and graded on accuracy and response time. Each task contained a different data used to derive the information presentation. After the tasks and questions regarding the tasks, the respondents were then asked a series of demographic questions.

The first survey group was given a survey instrument that included all the elements as explained in the preceding text as well as an interactive graphic DV and an interactive tabular DV. The second survey group was given a survey instrument similar to the first group.
A total of 675 survey responses were collected. Demographic information for respondents is as follows: 59% of the respondents were male and 40% were female. 33% of the respondents in the study were between the ages of 26 and 34. Interestingly, 24% of the respondents indicated that they had some experience with decision support systems (DSS). Survey development was accomplished by adapting scales from previously validated scales. The survey was piloted by an expert panel of IS faculty and Ph.D. students to ensure survey validation, and to gauge average

time to completion. Comments and suggestions by the expert panel were used in fashioning the final version of the survey, which was distributed to students.

Data Analysis and Results

Analysis was performed using SmartPLS, a partial least squares (PLS) analysis solution (Ringle, Wende, & Will, 2005). PLS is considered appropriate for studies early in the exploratory stage as with the current study (Ainuddin, Beamish, Hulland, & Rouse, 2007). Factor loadings are all above the recommended 0.7 threshold (Henseler, Ringle, & Sinkovics, 2009) with all values greater than 0.83 indicating good discriminate validity (Henseler et al., 2009).

Table 13

CFA Tabular Data

Scale Item	Mean	Std. Dev.	Cronbach's Alpha		Factor Loadings				
					1	2	3	4	5
TblEOU1	4.23	.883	0.938	0.957	**.839**	.172	-.068	-.132	.204
TblEOU1	4.26	.840			**.869**	.182	-.036	-.136	.144
TblEaOU3	4.29	.825			**.873**	.176	-.046	-.112	.152
TblEOU4	4.36	.782			**.793**	.029	-.078	-.080	.240
TblUseful1	4.25	.839	0.929		**.865**	.136	-.003	-.161	.186
TblUseful2	4.24	.839			**.875**	.160	-.005	-.153	.157
TblUseful3	4.06	.943			**.741**	.199	.148	-.153	.113
TblUseful4	4.26	.809			**.838**	.190	.026	-.116	.173
TblEmo3	2.59	1.070	0.923		-.064	.177	**.850**	.248	-.009
TblEmo4	2.49	1.113			-.009	.246	**.867**	.203	-.050
TblEmo5	2.46	1.096			-.058	.124	**.888**	.280	-.050
TblEmo6	2.77	1.103			.083	.346	**.770**	.126	-.029
TblDesign1	3.07	1.062	0.936		.165	**.873**	.196	.106	.038
TblDesign2	3.42	1.008			.235	**.741**	.121	.012	.062
TblDesign3	3.15	1.067			.151	**.859**	.231	.060	.056
TblDesign5	3.21	1.021			.193	**.894**	.148	.054	.020
TblDesign6	3.22	1.031			.171	**.885**	.172	.091	.046
TblCogLd1	2.31	1.087	0.905		-.207	.146	.214	**.831**	-.073
TblCogLd2	2.34	1.148			-.112	.044	.199	**.826**	-.088
TblCogLd4	2.25	1.126			-.220	.098	.199	**.829**	-.126
TblCogLd5	2.05	1.154			-.214	.024	.216	**.824**	-.131
TblTskConf1	4.40	.761	0.949		.530	.093	-.079	-.184	**.771**
TblTskConf2	4.40	.763			.489	.080	-.055	-.202	**.792**
TblTskConf3	4.41	.753			.432	.067	-.035	-.140	**.823**

Cronbach's alpha is a well-accepted measure of reliability. A minimum value of 0.7 for Cronbach's alpha is desired (Nunnally & Bernstein, 1994) with values less than 0.6 indicating a lack of internal validity. For this study Cronbach's alphas are all greater than 0.7 indicating satisfactory internal reliability (Henseler et al., 2009). Table 3 shows the descriptive statistics, loading factors, and Cronbach's alphas for the pretest data. Table 4 and 5 shows the descriptive statistics, loading factors, and Cronbach's alphas for the tabular and visual tasks.

Table 14

CFA Graphical Data

Scale Item	Mean	Std. Dev.	Cronbach's Alpha		Factor Loadings				
					1	2	3	4	5
VisEOU1	3.57	1.216	0.922	0.966	**.863**	.157	-.138	.043	.248
VisEOU2	3.56	1.209			**.879**	.167	-.057	.079	.179
VisEOU3	3.57	1.158			**.811**	.274	-.148	.106	.132
VisEOU4	3.88	.980			**.790**	.190	-.010	.078	.151
VisUseful1	3.86	1.103	0.948		**.876**	.160	-.131	.011	.252
VisUseful2	3.84	1.137			**.882**	.178	-.141	.007	.221
VisUseful3	3.71	1.098			**.784**	.323	-.125	.124	.122
VisUseful4	3.84	1.101			**.871**	.218	-.121	.056	.214
VisEmo3	2.73	1.074	0.888		-.008	.102	.249	**.806**	-.040
VisEmo4	2.43	1.138			.139	.208	.171	**.838**	.014
VisEmo5	2.49	1.143			.006	.132	.283	**.870**	-.013
VisEmo6	2.92	1.136			.224	.327	.113	**.714**	.060
VisDesign1	3.36	1.054	0.944		.195	**.863**	.026	.122	.057
VisDesign2	3.23	1.092			.224	**.809**	.060	.177	.077
VisDesign3	3.33	1.062			.247	**.869**	.026	.149	.043
VisDesign5	3.32	1.052			.239	**.870**	.013	.163	.116
VisDesign6	3.28	1.071			.258	**.862**	.034	.163	.071
VisCogLd1	2.59	1.078	0.868		-.139	.104	**.849**	.201	-.075
VisCogLd2	2.59	1.091			-.026	.031	**.788**	.205	-.040
VisCogLd4	2.51	1.094			-.184	.040	**.819**	.201	-.166
VisCogLd5	2.39	1.272			-.234	-.046	**.747**	.173	-.242
VisTskConf1	4.00	1.004	0.971		.420	.108	-.195	-.003	**.844**
VisTskConf2	4.03	.981			.430	.126	-.194	.010	**.841**
VisTskConf3	4.02	.971			.416	.121	-.197	-.002	**.849**

Table 15

CFA Pre-Test Data

Scale Item	Mean	Std. Dev.	Cronbach's Alpha	Factor Loadings				
				1	2	3	4	5
PreConf1	4.17	.871	0.930	.105	.267	**.869**	.218	-.015
PreConf2	4.2	.838		.124	.270	**.868**	.200	-.032
PreConf3	4.08	.884		.180	.240	**.874**	.132	.001
TaskKnow1	4.03	.884	0.937	.127	**.875**	.278	.127	.042
TaskKnow2	3.91	.936		.173	**.891**	.227	.128	.026
TaskKnow3	3.94	.951		.160	**.894**	.251	.106	.001
SoftEff1	2.62	.932	0.706	.114	.081	.009	.118	**.856**
SoftEff2	1.99	1.080		.141	-.034	-.044	-.041	**.875**
CompEff1	4.09	.857	0.915	.164	.147	.202	**.890**	.002
CompEff2	4.09	.857		.168	.157	.199	**.896**	-.007
CompEff3	3.83	.995		.168	.050	.111	**.877**	.105
TaskAtt1	3.90	.825	0.88	**.749**	.165	.107	.180	-.042
TaskAtt2	3.64	.854		**.856**	.064	.072	.111	.055
TaskAtt3	3.76	.880		**.775**	.187	.135	.068	.023
TaskInvlve1	3.56	.892		**.806**	.102	.082	.102	.164
TaskInvlve4	3.38	.972		**.802**	.013	.072	.131	.202

The following figures represent the core framework model of decision making in the presence of data visualizations. Figure 9 represents the interactive graphical data visualization model and its effects on the core constructs in the framework. Figure 10 represents the interactive tabular data visualization model and its effects on the core constructs in the framework. There are differences to be noted. In the interactive tabular model, the R-squared of task performance is 28.8% while, in the interactive graphical version, it is 56.2% indicating that using the interactive graphical DV positively influenced task performance outcomes more than its tabular counterpart. Also, the core constructs (usability, emotion, post task confidence, and cognitive load) all exhibited a higher r-squared as well as path coefficients in the graphical model as opposed to the tabular model, indicating that the graphical DV model is a better fit.

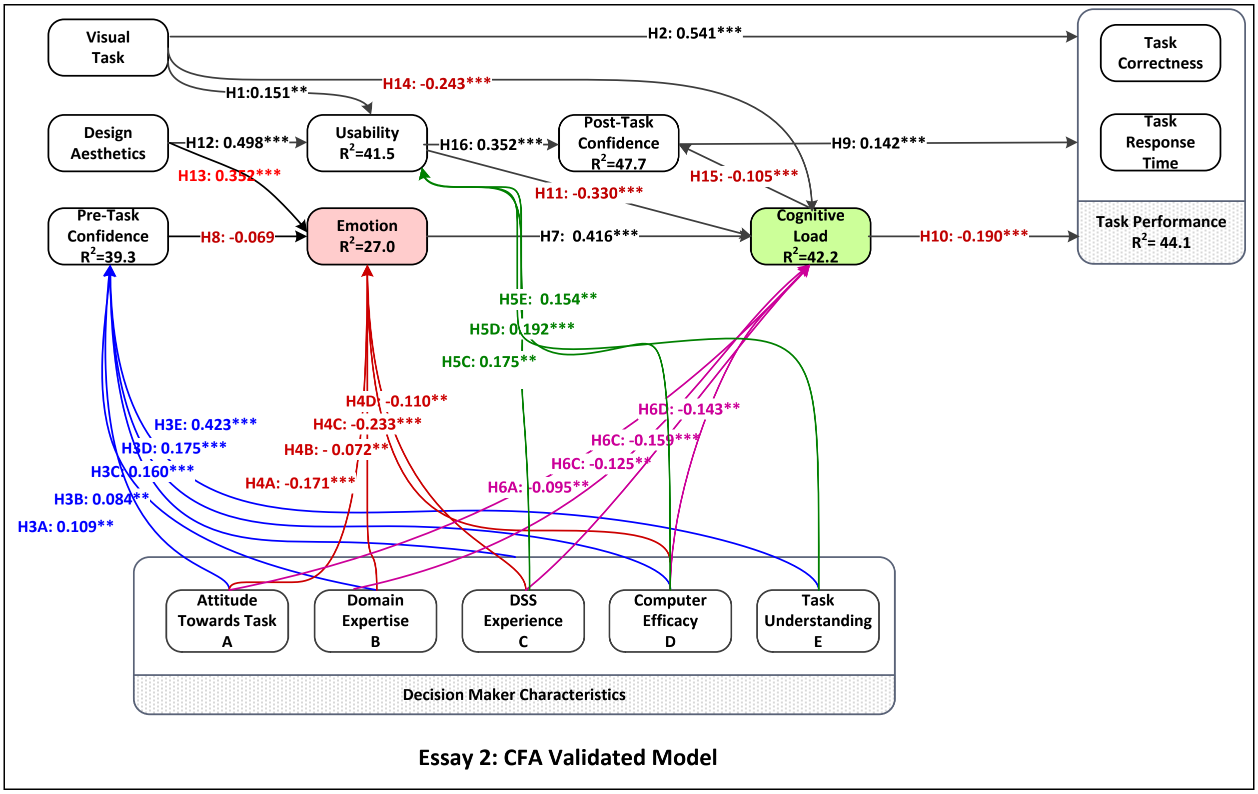


Figure 9. Structural model.

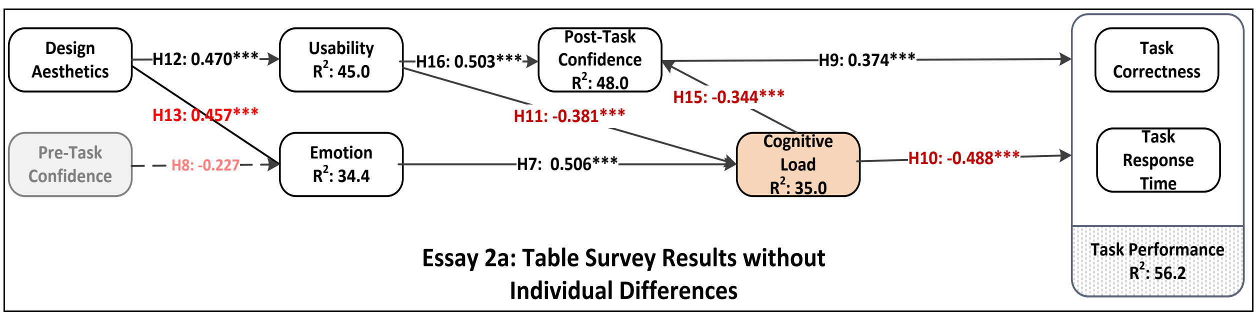


Figure 10. Model using interactive graphical DV data.

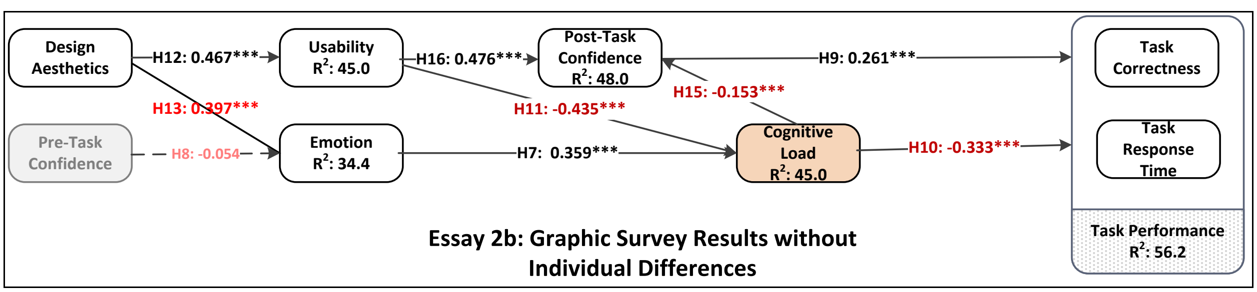


Figure 11. Model using interactive tabular DV data.

Construct Analysis

Table 16 shows the means for the constructs in the model related the interactive tabular task versus the interactive graphical task using the titanic data set. All construct mean differences are statistically significant. Compared to the tabular task, the graphical task measured better with regards to accuracy, usefulness, post task confidence, visual design, and emotional task engagement. The results showed an increase in task accuracy, and a reduction in task time and cognitive load, which suggests that respondents processed graphical DV more easily and that it provided a better task fit. Although all differences were statistically significant according to the levene's test, review of the mean charts indicate the largest magnitude of differences were task accuracy, time, and cognitive load with remaining differences being smaller.

Table 16

Mean Difference Analysis

	Static	Graphic
Decision Time	183	129
Decision Accuracy	83%	94%
Cognitive Load	2.84	2.18
Usefulness	3.81	4.18
Post Task Confidence	3.7	4.25
Visual Design	3.3	3.5
Emotion	3	2.7

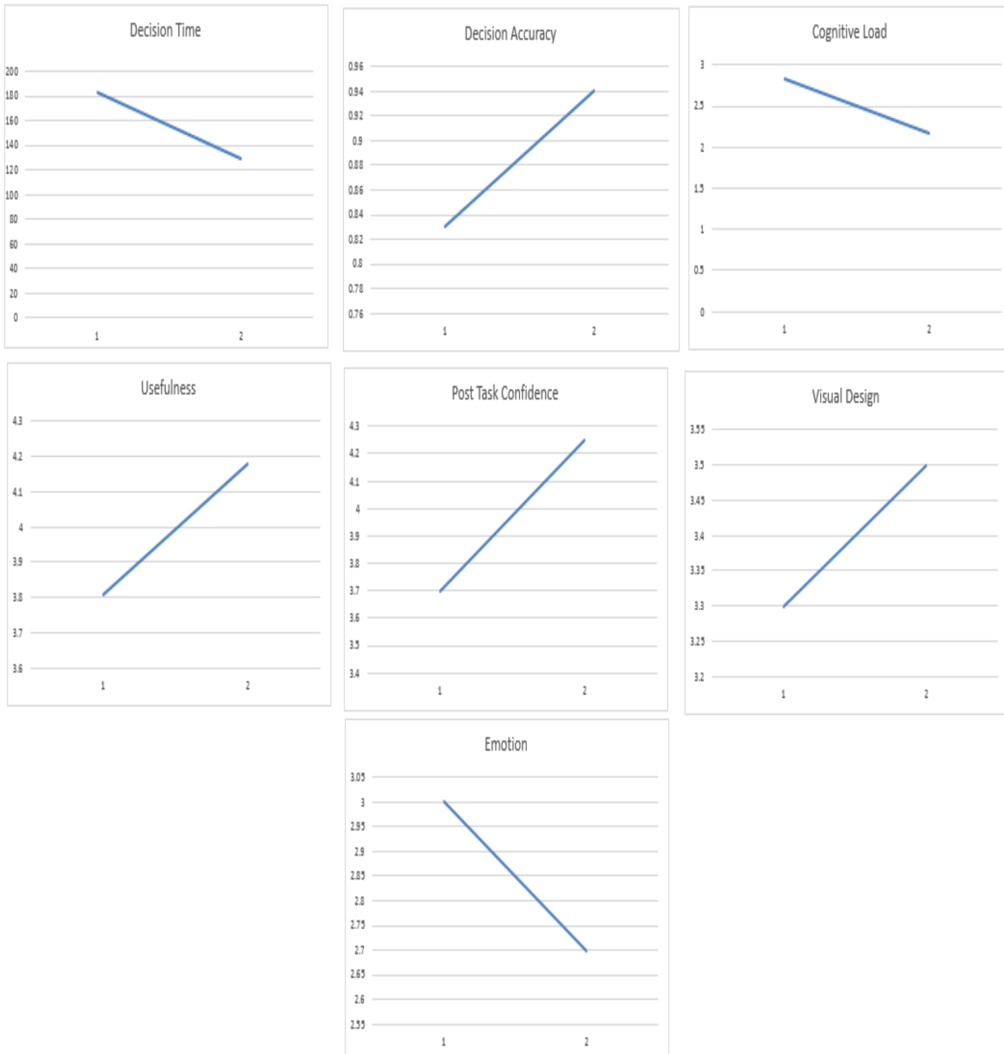


Figure 12. Mean charts.

Results

The structural model is shown in Figure 12. According to the data analysis cognitive load, task confidence, and visual task account for 50.9% of the total variance associated with task performance. Visual task, and design aesthetics together account for 41% of the total variance associated with the usability of the visual aid. Individual decision maker characteristics, specifically, educational level, DSS experience and computer efficacy account for 39% of the total variance associated with pre task confidence. Post task confidence, usability of the visual aid, emotion, and software efficacy account for 42% of the total variance associated with cognitive load. The usability of the visual aid accounted for 47% of the total variance associated with post task confidence. Lastly, design aesthetics, pre task confidence, domain expertise, task involvement, and software efficacy accounted for 27% of the total variance associated with the emotion responses elicited from the visual aids. The data emphasis the effectiveness and performance increase in the interactive graphical DV over the interactive tabular DV,

Table 17

Hypotheses Evaluation

	Hypothesis	Evaluation
H1	Information presentation in a visual format will have a positive effect on perceived usability	Supported
H2	Information presentation in a visual format will have a positive effect on task performance	Supported
H3(A-H)	Individual differences will have an effect on pre task confidence	Partially Supported
H4(A-H)	Individual differences will have an effect on Emotion	Partially Supported
H5(A-H)	Individual differences will have an effect on Usability	Partially Supported
H6(A-H)	Individual differences will have an effect on Cognitive Load	Partially Supported
H7	Emotion elicited by the information presentation will have a positive effect on cognitive load	Supported
H8	Pre Task confidence will have a negative effect on emotion	Not Supported
H9	Post Task confidence will have a positive effect on task performance	Supported
H10	Cognitive load will have a negative effect on task performance	Supported
H11	Perceived usability of the information presentation will have a negative effect on cognitive load	Supported
H12	Perceived design aesthetics will have a positive effect on perceived usability	Supported
H13	Perceived design aesthetics will have a negative effect on emotion	Supported
H14	Information presentation in a visual format will have a negative effect on cognitive load	Supported
H15	Cognitive Load will have a negative effect on Post Task Confidence	Supported
H16	Perceived usability will have a positive effect on post task confidence	Supported

Discussion and Conclusion

The goal of this research is to investigate factors that influence the decision making process and with it, task performance. In order to accomplish this, the study proposes and empirically tests a decision making success model based upon TAM, TTF, and cognitive fit in order to explain the dynamic and complex relationships involved in the decision making process. This study confirms the strong relationship between individual differences on emotion (H7), usability (H3), cognitive load (H5), and pre task confidence (H4, H15) and by extension, task performance. In addition, this study explores the role of design aesthetics, and visual presentation with respect to usability of information presentations and cognitive load, and finds that design aesthetics positively influence information presentation usability (H12), and emotion negatively (H13). Further, visual task representation also influence usability (H1) and overall task performance (H2). Results also indicate that cognitive load greatly negatively influences task performance (H10), as well as post task confidence (H9). The relationship between visual task representation and usability (H1), and task performance (H2), indicates that the information presented in a visual format was deemed more useful and translated into superior task performance.

Conclusions

This study extends the body of knowledge by exploring the many dimensions of decision making and the decision makers' role and influence in the process. This is one of the first studies to utilize interactive data visualization and explore their effect on task performance. Findings indicate that not only do individual differences effect task performance, but so do information presentation, attitudes, and design aesthetics.

This study also integrates the concepts of visual design and individual differences with TAM, TTF, and cognitive fit in order to explore their relationship to task performance. The relationships explored and empirically tested in this model indicate that task performance can be predicted and gives rise to the notion of better decision making process through inclusion of a multitude of dimensions, specifically those inherent to the decision maker themselves.

CHAPTER 6

STUDY 3 THE EFFECT OF INTERACTIVE VS STATIC DATA VISUALIZATIONS ON TASK PERFORMANCE

Introduction

Researchers first investigated task technology fit (TTF) (Dishaw & Strong, 1999; Goodhue & Thompson, 1995; Zigurs & Buckland, 1998) and computer information display (Dickson, 1977; Lucas, 1975) fifteen and forty years ago respectively, at a time when the incarnations, capacities, and uses of both computer hardware and software were very different from today. The aforementioned research remains authoritative, although its temporal separation from current practice suggests that an examination of these concepts with current technological capabilities is merited. Because diverse imagery is currently ubiquitous in data visualization (DV) and is available on practically every personal computing device, this research examines, in a contemporary end-user context, the TTF and tabular versus graphic image research.

The goal of this research is to address a TTF/DV gap in the IS literature by analyzing both the TTF and tabular versus graphic image research within the context of current DV technology. As it integrates recent literature, this research posits and tests a new model that extends TTF with user attitudes and perceptions about the tabular and graphic DV, and then measures effects on task performance.

This research contributes to TTF theory by extending it with attitudes and perceptions about the DV, developing and examining how the resulting model enhances overall fit and task performance. This research also contributes to the table versus graph research by assessing its relevance to the contemporary DV context. Finally, this research contributes by combining the two enhanced theories into a new, integrated TTF/DV task performance model, testing the model and explaining the significance of the results.

This study also investigates how users perceive and process tabular and graphical information using different levels of cognitive effort (Lindgaard, Fernandes, Dudek, & Brown, 2006; Tufte & Graves-Morris, 1983). Based upon our literature review we expect that decision makers expend more cognitive effort assimilating data depicted via tabular DV and less cognitive effort processing graphical DV (Daniel, 1992; W. P. Dickson, 2005; Feyyad, 1996; Robey & Farrow, 1982; Tufte & Graves-Morris, 1983). Therefore, this research investigates the role of cognitive load on user attitudes and perceptions about the task.

DV is an outgrowth of mainframe computer and personal computer (PC) text-based tables and graphs. The very first personal computer "killer apps" were spreadsheet/information display programs: VISICALC for the Apple II in1979 (Green, 1980) and Lotus 123 for the IBM PC in 1982 (Williams, 1982). Although users venerated both programs in their day, pervasive replacement of the DOS PC operation system (OS) with Windows 3.x in the early 1990s spelled their eventual demise, the applications eventually being overwhelmed by Microsoft's more graphically advanced Excel (Vaughan-Nichols, 2013). Often, mainframe terminal displays and VISICALC/Lotus 123 PC screens appeared remarkably similar; both media typically represented information textually and/or with simple graphs against a dark background (Figure 1).

When researchers published the original table versus graph research, (G. W. Dickson, 1977; Henry C Lucas, 1975), modern DV tools and prerequisite computer capabilities did not exist. Researchers conducted the seminal IS visual studies without a desktop OS or sophisticated DV tools. This early computing environment also denied users access to the many features of current DV.

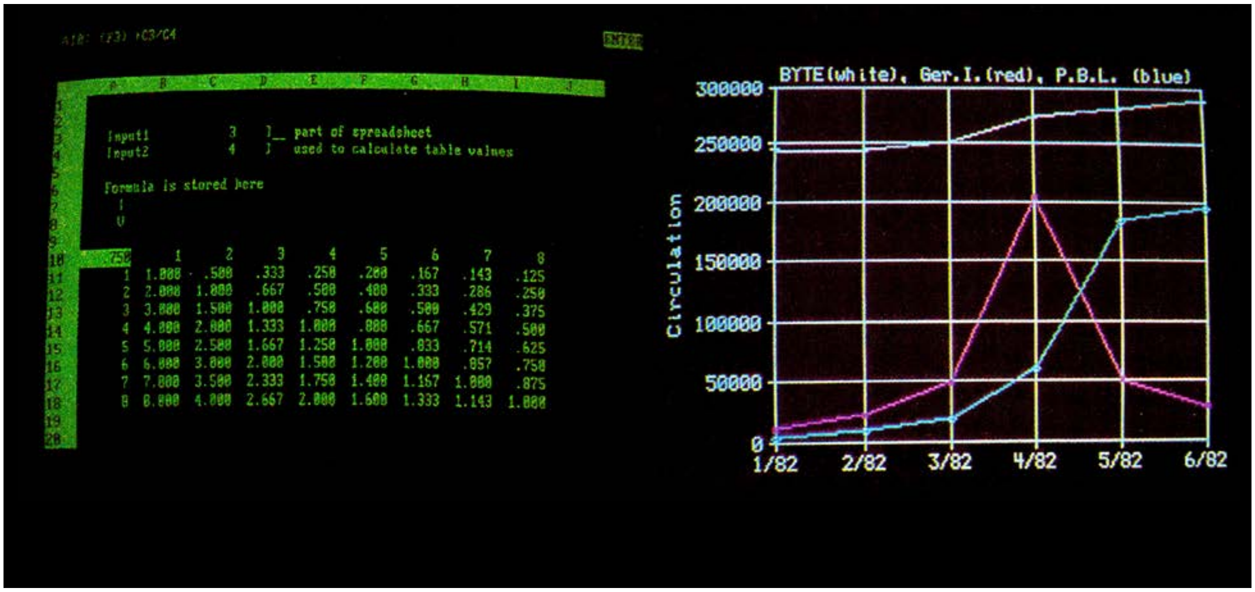


Figure 13. Traditional tabular versus graphics images.

Data visualization (DV) is an emerging and powerful tool that is capable of distilling the world's data, information, and knowledge into beautiful, interesting, and useful visualizations, graphics and diagrams (McCandless, 2013). Widely varied information imagery is ubiquitous in contemporaneous DV. In 2014, DV is available on practically every personal computing device, is distributed across most large cultural and socio-economic segments of the population, and is used across all age groups (Chen et al., 2012; Hargittai & Hsieh, 2012). Integrating DV into any decision making process can be best achieved when a fit exists between the DV technology and the DV information, in such a way that their intersection enhances task performance. Although the seminal TTF literature is more recent than the tabular versus graphic image research, it also merits revisiting in conjunction with DV in order to facilitate DV's growth in popularity and usage. Responding accordingly, this research proposes an extended theoretical TTF/DV task performance framework and tests effects of the resulting new model on task performance.

The remainder of this work is organized as follows. The theoretical foundation section explains the underlying concepts and theory of this research, after which the model development and hypotheses section details the research constructs and their place in the decision making process framework, along with the hypothesized relationships between the constructs. The survey development section describes the how the authors have adapted the construct items and incorporated them into the survey, as well as the treatment preparations and pilot testing processes. The methodology section describes the research methods used to empirically test the research model. The preliminary data analysis section presents the descriptive statistics of the study and describes the respondent demographics. The split sample method section describes how the authors subdivided the large data sample for exploratory factor analysis (EFA) and confirmatory factor analysis (CFA) purposes. The withheld sample EFA data analysis section discusses the authors' analysis of the first half of the split sample, presenting the results of the hypothesis testing, followed by the verification sample CFA data analysis section, which discusses the confirmatory results of testing the second half of the sample. The group differences analysis section details the resulting differences between the two treatment groups, after which the summary discussion section reviews and summarizes the study findings. The study's final four sections are conclusion, implications of the study, research limitations, and suggestions for future research.

Theoretical Foundation and Model

Researchers have employed the three TTF models which includes a utilization focus, a fit focus, and a combination utilization and fit focus. These TTF models link technology with performance (Goodhue & Thompson, 1995) to study a diverse range of information systems, some whose attributes include user attitudes and perceptions (Mathieson & Keil, 1998). In addition, researchers have combined the TTF models with other models related to IS outcomes, such as the technology acceptance model (TAM) (Dishaw & Strong, 1999). This research utilizes the third TTF model (TTF3), which combines utilization and fit, to describe concepts that enhance performance (Table 1, Column 1). This work then extends TTF3 by incorporating tabular versus graphic DV task stimuli into the model.

The resulting TTF/DV model extends TTF3 (Table 1, Column 2). It operationalizes task characteristics with the task survey questions, technology characteristics as the table versus graph visual stimuli as they appear on the computer, precursors of utilization constructs (usability, emotion, user attitudes, and design aesthetics), utilization, which is the DV context of the experiment, and finally performance impact constructs (task performance, post-task confidence, and cognitive load).

Table 18

Extension of the TTF3 Model Concepts into a Testable TTF/DV Model

TTF3: Utility and Fit Combination (Goodhue & Thompson, 1995) Model Concepts	Extended TTF3: TTF/DV Task Performance Model Model Implementation
Task characteristics	Task survey questions
Technology characteristics	Computer visual task stimulus: table
	Computer visual task stimulus: graph
Precursors of utilization	Usability construct
	Emotion construct
	Attitudes construct
	Design aesthetic construct
Utilization	The DV context of the experiment
Performance impacts	Task performance construct
	Cognitive load construct
	Post task confidence construct

The present model integrates these extensions via inclusion of visual design, cognitive fit, task confidence, and usability to support its premise: user acceptance and sustained engagement with the DV will promote task performance, and therefore the TTF. Combining these theories provides insight into the multifaceted relationship that drives the successful fit of a DV information system and yields understanding of individual and organizational impact on decision-making performance. This examination and extension yields a new TTF/DV task performance model which we test in a survey experiment. The aforementioned research remains authoritative, although its temporal separation from current technology suggests that an examination of these concepts is merited.

Research Question

DV is available on practically every personal computing device, is distributed across most large cultural and socio-economic segments of the population, and is used across all age groups. Given the ubiquitous nature of the current DV technology, integration of contemporary literature with the TTF literature and tabular versus graphic image research is a worthwhile endeavor. Thus, this research proposes the following research question:

By extending TTF to embrace user attitudes and perceptions of the contemporary DV task and its environment, does the extended TTF model result in better decision-making and task performance?

Data Visualization

The manner of information representation exerts a powerful influence on user attitudes and task performance (Buja, Cook, & Swayne, 1996; Lurie & Mason, 2007; Tukey, 1990; Ware, 2012). From an IS perspective, contemporary DV is a process that visually represents data by using applied visual design theory, techniques, and tools to reveal meaningful trends and patterns in data (Cawthon & Moere, 2007; Cyr, Head, Larios, & Pan, 2009; Speier & Morris, 2003). From a designer's perspective, DV is a creative tool that synthesizes data, information, and knowledge into informative, inspiring, and aesthetically pleasing visualizations, graphics and diagrams (McCandless, 2013). By presenting the survey questions in a contemporary DV context, this research draws from both perspectives as it measures tabular versus graphic task performance with the new TTF/DV model.

Graphical images are instruments for reasoning about quantitative information, however of all the available methods for analyzing and communicating statistical information, well-designed data graphics are often the simplest and the most powerful (Tufte & Graves-Morris, 1983). Graphical DV exploits natural human visual acuity, allowing users to grasp information quickly and accurately (Lindgaard et al., 2006; Lurie & Mason, 2007). In DV tasks, for users to evaluate the functionality of the DV requires several seconds, while perception of DV appearance is virtually instantaneous, so that users effortlessly form impressions and quickly bond with the design (Bloch, 1995; Daniel, 1992; Kelton, Pennington, & Tuttle, 2010). Ostensibly anticipating benefits of DV, current researchers are studying less the functional and utilitarian aspects of decision making and have focused more recently on user visual effects and experiences (Cyr et al., 2010; Hassenzahl & Tractinsky, 2006). Another motivation for graphical DV research is the discovery that pre-cognitive, visual perceptions have long-lasting effects (Bloch, 1995). In addition, the recent introduction of frameworks for examining aesthetics in the IT domain indicates this research is timely (Hassenzahl & Tractinsky, 2006) The study examines whether visual presentation plays a critical role in creating a fit, decision-making, and task performance by examining decision scenarios delivered with tabular and graphical DV stimuli.

Cognitive Fit

Cognitive fit occurs in problem solving when the problem representation and any tool or aids used in the task support the processes required to perform the task (Iris Vessey, 1994). cognitive fit theory (CFT) provides a theoretical basis for understanding how information formatting supports decision-making tasks (Vessey, 1991). A cognitive fit, or mental match, occurs when the information accentuated by one particular presentation format matches the information required for a user to most easily complete the task

The cognitive fit focus has been most evident in graphs versus tables research (Benbasat & Dexter, 1985; Dickson, DeSanctis, & McBride, 1986), which indicates the influence of data representation on performance depends on "fit" with the task, thus supporting cognitive fit and task performance (Jarvenpaa, 1989; Vessey & Galletta, 1991). Lurie and Mason (2007) concluded that data visualizations can improve efficiencies in routine tasks and enable users to uncover new insights that would have otherwise been unnoticed. Cognitive load, which is a component of cognitive fit plays an important role in this study as we measure the effect of the DV tasks cognitive load on task performance. It is our aim to examine the TTF, with the added component of cognitive load for a more comprehensive "fit."

Task Performance

The linkage between information technology and individual performance has been an ongoing concern in IS research. This article presents and tests a new, comprehensive model of this linkage. Task performance is the outcome variable in the TTF model and is consistent with this study's aim to examine the precursors to task performance utilizing TTF. In the context of this research, task performance is a vital outcome variable that assesses how well a task was performed, conditioned by the overall effectiveness of the DV (Heath & Etheridge, 1991; Shneiderman, 1996; Tory, Kirkpatrick, Atkins, & Moller, 2006). Analogous to early DV research, most task performance research that dealt with visualization either centered on the technical aspects of producing the visualizations (Heath & Etheridge, 1991; Singh, Gupta, & Levoy, 1994) or on assessing the more functional visualizations that predominated in the 1970s and 1980s (Ghai, 1981; Lucas, 1981; Tufte & Graves-Morris, 1983). Our study contributes to task performance DV-related research in the post-millennial context by employing a contemporary, emerging DV tool (JMP by IBM SAS) to create our survey images.

This approach is supported by research indicating that DV components of business intelligence (BI) increase usability and decision making effectiveness, which in turn increases the rapidity and accuracy of task performance (Cawthon & Moere, 2007). This holistic view of the task performance and decision-making has yet to be fully explored. One reason is that the technology of prior research is dated and constrained by the technology of the time, including limitations in display, software, and processing capability (Benbasat & Dexter, 1985; DeSanctis & Gallupe, 1984; Jarvenpaa, 1989). This research updates previous task performance, TTF/DV and decision making research (Schmell & Umanath, 1988; Washburne, 1927; Watson & Driver, 1983) with current technology and by enhancing prior research with user attitudes and perceptions about the DV, filling a gap in the literature with fresh perspectives for a new generation of users.

Task Technology Fit (TTF)

TTF is the degree to which a technology assists an individual in performing his or her tasks. More specifically, TTF is the correspondence between task requirements, individual abilities, and the functionality of the technology (Goodhue & Thompson, 1995). MIS literature has examined TTF many times, for example, research on data representation and visualization such as tables, and graphs has concluded that the best representations depends on task requirements (Tan & Benbasat, 1990). This "fit" focus has been most evident in research on the impact of graphs versus tables on individual decision-making performance. Two studies report that over a series of laboratory experiments, the impact of data representation on performance depended on the fit with the task (Benbasat et al., 1986; Dickson et al., 1986). Another study proposes that mismatches between data representations and tasks slows decision making performance by requiring additional translations between data representations or increased cognitive load (Vessey, 1991).

TTF can account for enhanced task performance and improve system effectiveness by enabling a fit between the technology and the task itself (Zigurs & Buckland, 1998). Different types of visualization (e.g., 2d, 3d, mixed) can be effective for different types of tasks indicating that there is a “fit” between the task and the visualization that leads to optimal task outcomes (Tory et al., 2006). TTF becomes more important in today’s computing environment due to the ubiquitous nature of computer usage at the individual and organization level. Early DV research has shown the importance of graphs but fails to emphasize the high level of convenience we have today with modern graphic visualizations, graphics, and diagrams.

This research contributes to TTF by adding attitudes and perceptions about the DV and how they enhance the overall fit and task performance and by examining TTF’s relevance in a contemporary DV task environment. Therefore the focus of this paper is the use of task technology fit within the realm of data visualizations.

Information Presentation

This research examines the importance of presentation in tabular versus graphic image research for achieving optimal fit, this research tests the new TTF/DV model, which embraces user attitudes and perceptions of the contemporary DV task and its environment. In visual computer aided decision making, the decision maker visually perceives and cognitively processes the revealed outcomes to generate informed business decisions. The information systems that support DV concepts and methods contain business-oriented data visualization objects, such as dashboards, OLAP cubes, and balanced scorecards, that facilitate decision-making (Watson & Wixom, 2007). Even basic computers and hand-held devices are vastly more powerful and capable of displaying complex graphics and animations. Now more than ever, developers need to understand the value of DV and the process of creating them—how to blend visual form and system functionality to provide fresh sensory insights into massive, complex data (Buja et al., 1996; Cawthon & Moere, 2007; Healey, 1996).

Researchers have jointly studied task fit and information presentation, specifically regarding the effects of tabular versus graphical data representations and their effects on decision-making performance (Benbasat & Dexter, 1985; Ghai, 1981; Kelton, Pennington, & Tuttle, 2010; Vessey, 1991). Early on, Kolata (1982) recognized that graphic displays enable scientists to utilize the uniquely human ability to recognize meaningful patterns in the data, and to see patterns that might otherwise have been imperceptible. Vessey (1991) proposed cognitive fit as a response to the graph versus table research stream and proposes that the relationship between task and information presentation format lead to increased task performance for individual users as a result of this fit. Accordingly, this research utilizes components of cognitive fit to identify the preferred fit with the DV presented in the experiment and how it relates to task performance, as measured by the rapidity and task accuracy.

Visualization is necessary to explore and analyze large amounts of data. The advantage of data visualization is that the user is directly involved in the mining process through interactive visualizations (Keim, 2002). Interactive visualization allow for easier navigation and exploration of data, as well as hypotheses testing, something that is more time consuming to do with static visualization and increasingly difficult to do with vast amounts of data (Grinstein & Ward, 2002). This study utilizes interactive data visualizations and explores their influence on task performance and the decision making process. A majority of the interactive data visualization research deals with creating the DV (Buja et al., 1991; Ward et al., 2010). It is the aim of this research to investigate the effects of interactive visualizations on task performance in the context of an individual decision making process. Data analysis is fundamentally an iterative process in which you submit a query, receive a response, formulate the next query based on the response, and repeat. You usually don't issue a single, perfectly chosen query and get the information you want from a database. Therefore the purpose of data analysis is to extract unknown information, and in most situations, there is no one perfect query. People naturally start by asking broad, big-picture questions and then continually refine their questions based on feedback and domain knowledge (Hellerstein et al., 1999). This research explores the way in which users can utilize the interactive visualization to explore the data and answer specific questions regarding the dataset, that we hypothesize would be more difficult and time consuming given a static non interactive data visualization. As the extant research suggests, utilizing an interactive will allow for faster data discovery and querying that should enhance time to completion and accuracy.

Model Development and Hypotheses

This paper is the first to develop and test a theoretical model that encompasses multiple variables to address DV and their effects on individual task performance. This experimental model aims to extend this current DV research and examine task performance in a visual aided decision making environment.

- The Cognitive Load Construct

Cognitive load, a key component of cognitive fit theory and a factor in the proposed TTF/DV model, is defined and measured as mental effort on the part of the decision maker (Van Gerven, Paas, Van Merriënboer, & Schmidt, 2004). Cognitive fit asserts that for task success, there should exist an ideal fit between task information presentation and task requirements. If a fit does not exist, then the task becomes more difficult and the cognitive load becomes greater. Copious research indicates that choices of graphic versus tabular information presentation significantly influence decision outcomes (Dickson et al., 1986; Jarvenpaa, 1989; Tukey, 1990; I. Vessey & Galletta, 1991; Washburne, 1927). For this study, we adopt the premise from the aforementioned research that while tabular representations are more effective for some tasks, they may not be for others. For this research , the cognitive load construct is an indicator of TTF: decision makers dissipate more cognitive effort assimilating data depicted via tabular representations (Daniel, 1992; Dickson, 2005; Feyyad, 1996; Robey & Farrow, 1982; Tufte & Graves-Morris, 1983), while they dissipate less cognitive effort processing graphic DV (Lindgaard et al., 2006; Peak, Prybutok, Gibson, & Xu, 2012).

H1: Information presentation in a visual format will have a negative effect on cognitive load

H4: Emotion elicited by the information presentation will have a positive effect on cognitive load

H6: Cognitive load will have a negative effect on task performance

H7: Perceived usability of the information presentation will have a negative effect on cognitive load

H10: Information presentation in a visual format will have a negative effect on cognitive load

H11: Cognitive load will have a negative effect on post task confidence

- The Visual Design and Emotion Constructs

Although marketing and web design have examined visual design and resulting emotional response to the design for decades (Cyr et al., 2010; Cyr et al., 2009; Cyr, Kindra, & Dash, 2008; Lindgaard et al., 2006; Salisbury, Pearson, Pearson, & Miller, 2001), little research in IS and TTF investigates this area.

Tractinsky (2004) argues that aesthetic design plays a major role in private, social, and business situations, positing that aesthetic design plays a key role in IT. Cawthon & Moere (2007) assert that aesthetics is a mechanism to explain and predict IS success, usability, and trust and to aid in the decision making process. In this study evidence suggests that across multiple visualization types, information presentations that were measured as being more visually appealing, scored higher on usability, and produced more correct responses. Much of the research into aesthetic design centers itself on website design and the user's perception of the site due to its perceived aesthetic value. These studies have shown that users respond positively to a website or interface perceived as aesthetically pleasing (Cyr et al., 2010; Cyr et al., 2008; Lindgaard et al., 2006). This research builds upon this stream by incorporating the visual design construct into our model framework; aesthetical perception of the DV design has been shown to influence the users' attitude, feelings of trust, and emotion, as well as its positive effects on fit. Emotional responses to visuals and interfaces are equally as important as aesthetic value in influencing user's attitudes and feelings toward a system or visual aid. Beaudry and Pinsonneault (2010) found that emotions impact appropriate IS usage in work settings (Lindgaard et al., 2006). Other studies have found that users reactions to visuals are not only immediate but also invoke users to feel emotions in the form of sensations, pleasure and arousal (Osborne, 1979) indicating that users reactions to the medium does influence their attitudes towards the system. Because emotion, visual design, and fit are related, this research builds upon previous emotion response research by incorporating the emotion response construct into our model framework.

H2: Information presentation in a visual format will have a positive effect on task performance

H8: Perceived visual design will have a positive effect on perceived usability

H9: Perceived visual design will have a positive effect on emotion

- The Usability Construct

Usability is a cornerstone of the technology acceptance model (TAM) by determining whether an IT system is successful by being both easy to use and useful (Davis, 1989). This research applies TAM to TTF and tabular versus graphic data representation. These systems facilitate the decision making process must also be deemed both easy to use and useful in order to be effective, and enable positive task outcomes (Doll, Hendrickson, & Deng, 1998). This research builds upon previous usability research by incorporating the usability construct into our model framework.

H3: Attitude towards task will have a positive effect on usability

H12: Perceived usability will have a negative effect on post task confidence

- Decision Confidence and Task Performance

Decision confidence has been widely studied going back decades in IS research. DeLone and McLean (1992) explored user confidence as one of the dependent variables responsible for information system success. Other studies conducted by Chervany and Dickson (1974) as well as Taylor (1975) expressed the importance of user confidence in an information system to ensure successful outcomes. A user's confidence in the system aids the user in the buy in of the system, and translates to a better understanding of the system, the task required, and ultimately a better decision can be made because of this confidence. Another study conducted by (Sharda, Barr, & MCDonnell, 1988) examined users of a decision support system (DSS) and how users were more confident when using a computer aided decision system than completing the tasks in a more traditional manner. The users that utilized the DSS were more confident in their decision and ultimately made better decisions. Therefore it is vital that we establish the dimensions which affect user confidence in their decision to better understand and equip them with decision making effectiveness.

H5: Post task confidence will have a positive effect on task performance

- Decision Making Process Framework

This research focuses on TTF and tabular versus graphic image research to propose a new TTF/DV framework and test the resulting TTF/DV task performance model. In developing this framework we grouped constructs into three major conceptual categories: task attributes, task perceptions, and task outcomes (Figure 14).

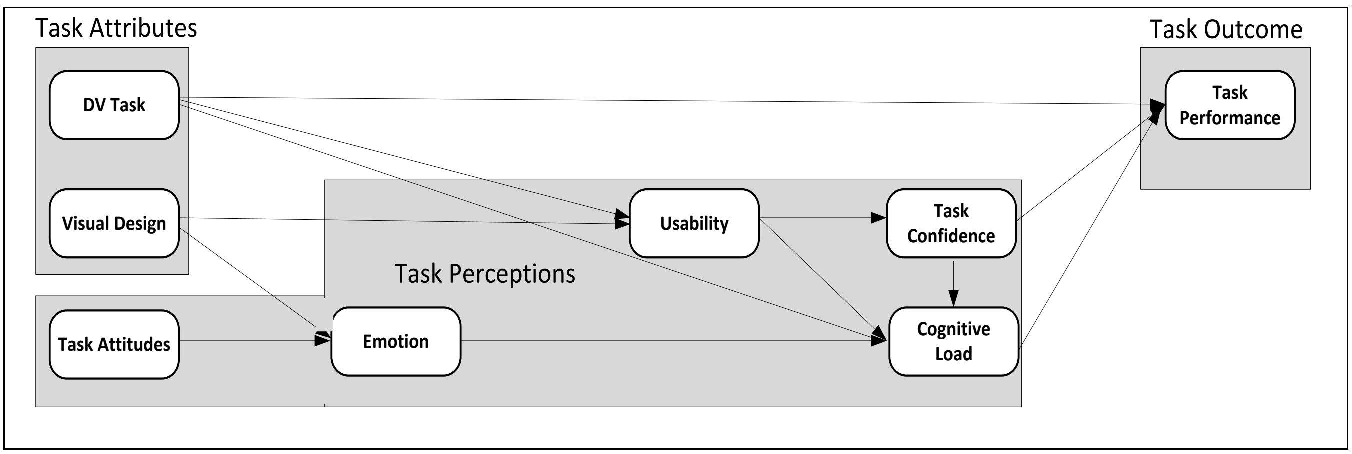


Figure 14. The decision making process framework.

Task attributes, the first conceptual category, includes the constructs DV task (tabular or graphical) and visual design of the DV. Task perceptions, the second conceptual category, includes the constructs pre-task confidence, emotion, usability, cognitive load, and post-task confidence. Task outcomes, the third and final category, includes the construct task performance. Task outcome is our dependent variable and in this study includes task performance, which is the percentage of correctly answered task questions. During the preliminary model development and pilot testing, results indicated task completion time to be insignificant, which is not surprising, as we designed the DV representations to be equivalent tabularly and graphically. Therefore, this research uses task accuracy as its sole measure of task performance.

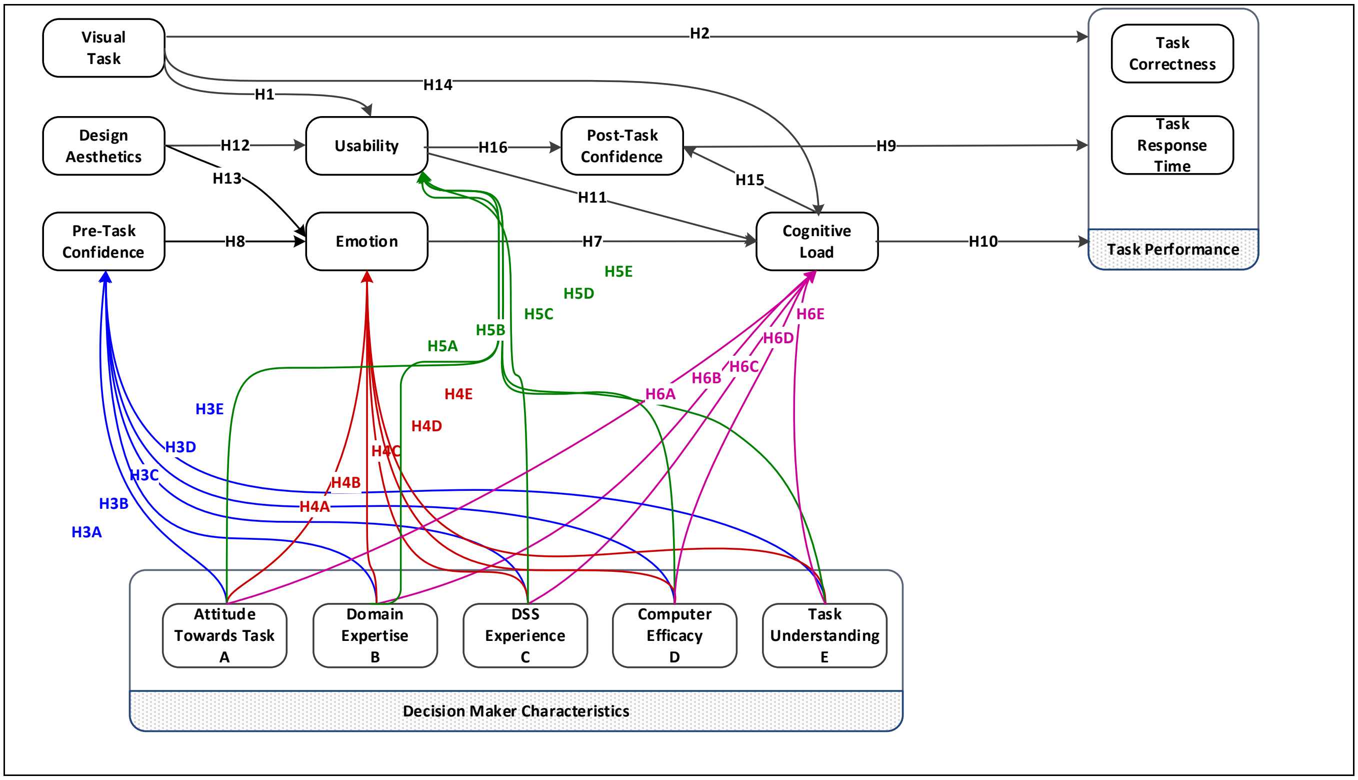


Figure 15. The testable TTF/DV model.

Survey Development

The authors developed the survey by adapting items from previously validated research constructs. They piloted the preliminary version using an expert panel of IS faculty and doctoral students (IT, decision science, marketing) to validate the instrument and to gauge average completion time. Resulting comments and suggestions by the expert panel helped refine the final version of the survey, which the authors then distributed to students (please see Appendix A for the construct table). The chosen DV tool was JMP by IBM SAS (SAS Publishing, 2012) to prepare and create the visuals for the survey, in recognition that JMP is a rapidly expanding tool with increasing industry adoption (Gartner Group 2013). Gartner places DV at the peak of its 2013 hype cycle and places JMP by IBM SAS at position number 4 in the top five most used

DV software, with 8% reported use and 35$ million in estimated revenue (Gartner 2013). During the pilot process, the authors achieved group agreement of information equivalency between the paired visual tasks. Both the researchers and the individuals involved in the pilot process agreed that the experimental tasks were appropriate for examining effects of DV on TTF and on cognitive load.

Methodology

We obtained a sample of 823 undergraduate and graduate students at a large university in the southwestern United States that was split into two randomly selected halves for the purposes of model development and model testing. The respondents completed a survey in one of two randomly assigned groups administered online through qualitrics survey administration platform. Each survey consisted of 125 questions: 23 demographic questions, 45 attitude and perception questions for the tabular task and 6 tabular task performance questions and 45 attitude and perception questions for the graphical DV task and 6 graphical task performance questions which were graded for accuracy and timeliness. Each survey group responded to two tasks representing equivalent information, one graphical DV and one tabular DV. We utilized a 2 X 2 survey design where task 1 contained information about mobile operating systems market share, while task 2 contained information about Titanic shipwreck survivors. This research design enables exploration of differences across the groups, and mitigates effects of the information presentation styles on the respondents. Respondents were assigned randomly to 1 of the 2 survey groups. The order in which the users were shown the visual and functional aids was randomized to mitigate order bias and increase reliability and validity of the sample. Group 1 was given an interactive graphical task and a static tabular task, while group 2 was given an interactive tabular task and a static graphical task. This was done so that we could examine the differences between the interactive and static DV aids.

Data Analysis and Results

We performed a preliminary data analysis to assess the descriptive statistics in our study and to examine the demographic information of our respondents. Demographic information for all respondents is shown in Table 2. 58% of the respondents were male and 41% were female. 77.2% of the respondents in the study were between the ages of 18 and 25. Interestingly, 18% of the respondents indicated that they had some experience with decision support systems (DSS) and have at least had two statistics courses and either an operations research or operations management course.

CFA Data Analysis

Factor loadings are all above the recommended 0.7 threshold (Henseler et al., 2009) with all values greater than 0.83 indicating good discriminate validity (Henseler et al., 2009). Nunnally & Bernstein (1991) recommend a Cronbach alpha minimum of 0.7 with values less than 0.6 indicating a reduction in internal validity. In this study, all Cronbach's alphas are greater than 0.7, indicating satisfactory internal reliability (Henseler et al., 2009). Table 4 shows the descriptive statistics, loading factors, and Cronbach's alphas for the tabular DV and graphical DV task as well as task perceptions. The CFA analysis confirmed the EFA analysis and chosen construct measures in both analyses were all above the recommended thresholds.

CFA Discussion

Analysis and results through PLS as reference in Figure 15, indicate that the CFA models supports our hypotheses. In the CFA model Task performance has an r-squared of 57.9% which is derived from graphical DV task, post task confidence, and cognitive load. Together, graphical

DV task, and visual design account for nearly 41% of the variance associated with usability. cognitive load is influenced by usability, graphical DV task, and emotion which combine to account for 42% of the variance explained.

Table 19

Verification Dataset (CFA) Statistics and Factor Analysis

Constructs	Std. Loading	Cronbach Alpha	AVE	Composite Reliability	Factor 1	2	3	4	5	6	7	8	9	10	11
VisEasyToUse1	0.98	0.951	0.873	0.955	.853										
VisEasyToUse2	0.97				.824										
VisEasyToUse3	0.98				.872										
VisEasyToUse4	0.97				.859										
VisUseful1	0.94	0.952	0.876	0.957	.869										
VisUseful2	0.94				.880										
VisUseful3	0.87				.796										
VisUseful4	0.96				.866										
VisEmotion3	0.84	0.886	0.748	0.92							.762				
VisEmotion4	0.89										.762				
VisEmotion5	0.91										.797				
VisEmotion6	0.80										.632				
VisDesign1	0.88	0.946	0.823	0.955			.804								
VisDesign2	0.87						.804								
VisDesign3	0.92						.858								
VisDesign5	0.93						.865								
VisDesign6	0.92						.863								
VisTaskCogLoad1	0.82	0.870	0.718	0.920						.793					
VisTaskCogLoad2	0.79									.825					
VisTaskCogLoad4	0.81									.726					
VisTaskCogLoad5	0.85									.711					
VisPostTaskConf1	0.97	0.967	0.939	0.976									.799		
VisPostTaskConf2	0.97												.803		
VisPostTaskConf3	0.97												.776		
TblEasyToUse1	0.92	0..942	0.852	0.957		.811									
TblEasyToUse2	0.94					.842									
TblEasyToUse3	0.93					.839									
TblEasyToUse4	0.89					.763									
TblUseful1	0.92	0.922	0.824	0.952		.841									
TblUseful2	0.94					.844									
TblUseful3	0.83					.734									
TblUseful4	0.92					.822									
TblEmotion3	0.90	0.822	0.813	0.947					.824						
TblEmotion4	0.93								.784						
TblEmotion5	0.93								.838						
TblEmotion6	0.86								.745						
TblDesign1	0.92	0.812	0.821	0.953				.839							
TblDesign2	0.81							.700							
TblDesign3	0.91							.838							
TblDesign5	0.93							.875							
TblDesign6	0.93							.855							
TblTaskCogLoad1	0.86	0.612	0.720	0.852								.710			
TblTaskCogLoad2	0.82											.631			
TblTaskCogLoad4	0.87											.775			
TblTaskCogLoad5	0.86											.724			
TblPostTaskConf1	0.96	0.894	0.894	0.97											.728
TblPostTaskConf2	0.96														.715
TblPostTaskConf3	0.91														.769
TaskAttitude1	0.78	0.895	0.712	0.912										.850	
TaskAttitude2	0.85													.813	
TaskAttitude3	0.80													.838	

CFA Hypotheses Results

This study explores the role of visual design, and visual presentation with respect to usability of information presentations and cognitive load, and finds that visual design positively influence usability (H8), as well as emotion (H9). Further, graphical DV task representation also influence usability (H1) and overall task performance (H2). Results also indicate that cognitive load negatively influences task performance (H6), as well as post task confidence (H11). Interestingly, the relationship between graphical DV task and usability (H1), and task performance (H2), and cognitive load (H10) exhibited a strong and significant relationship indicating that the graphical DV task allowed for higher task performance, usability, and lower cognitive load.

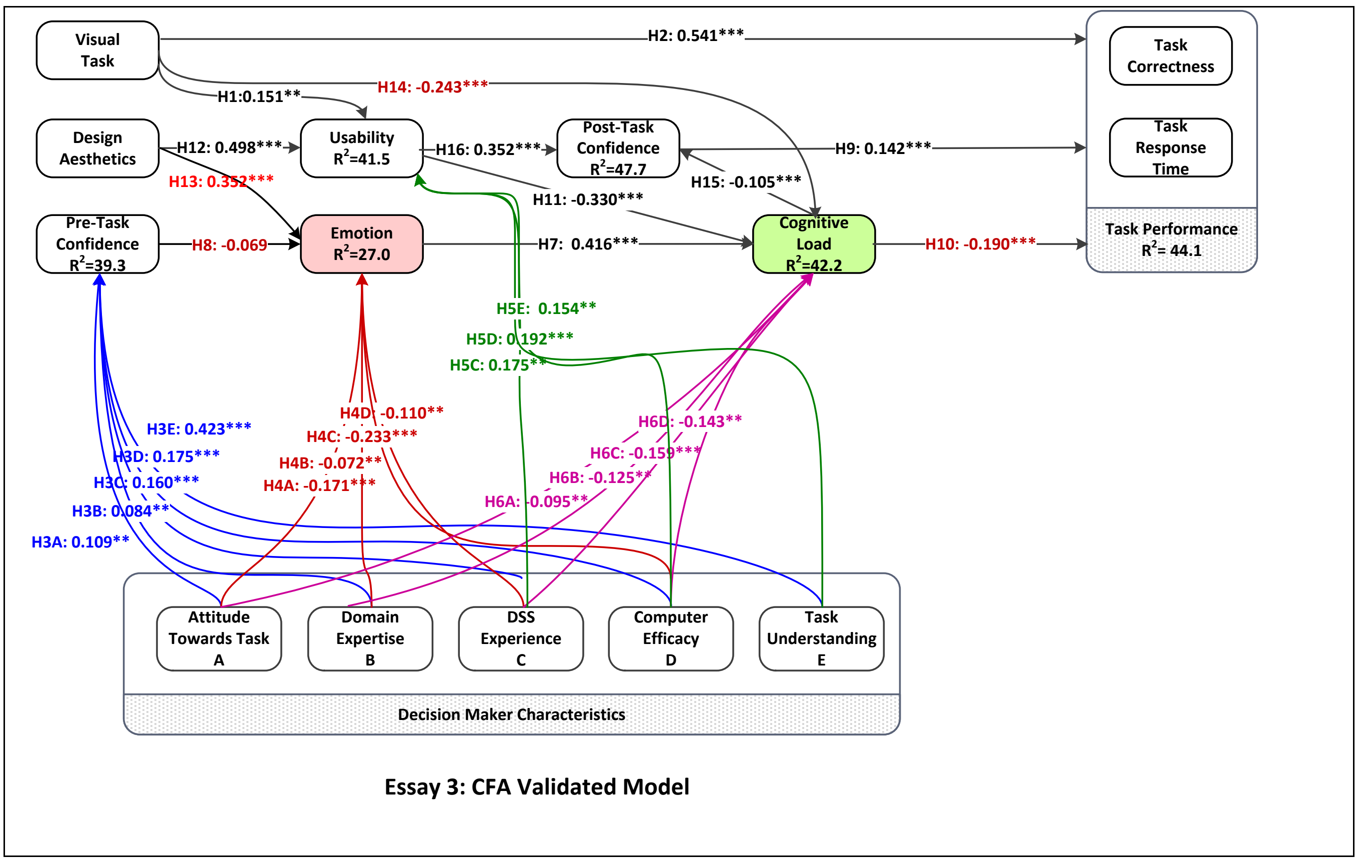
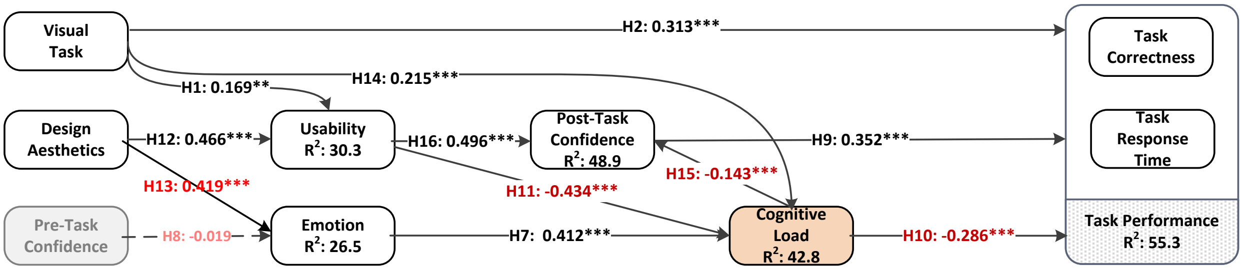
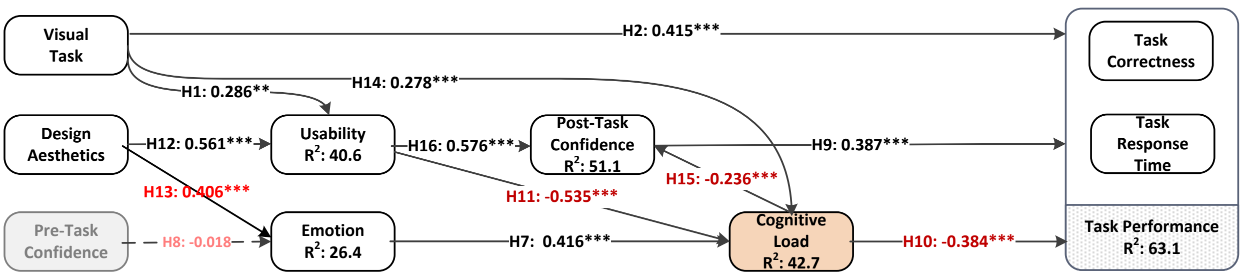


Figure 16. (CFA) structural model.



Essay 3a: Table Survey Results without Individual Differences

Figure 17. Table survey results without individual differences.



Essay 3b: Graphic Survey Results without Individual Differences

Figure 18. Graphic survey results without individual differences.

Construct Analysis

Table 23 shows the means for the constructs in the model related the tabular task versus the graphical task using the titanic data set. All construct mean differences are statistically significant. Compared to the tabular task, the graphical task measured better with regards to accuracy, usefulness, post task confidence, visual design, and emotional task engagement. The results showed an increase in task accuracy, and a reduction in task time and cognitive load, which suggests that respondents processed graphical DV more easily and that it provided a better task fit. Although all differences were statistically significant according to the Levene's test, review of the mean charts indicate the largest magnitude of differences were task accuracy, time, and cognitive load with remaining differences being smaller. Table 6A shows the means for each of the major constructs.

Table 20

Mean Differences Analysis

Mean Difference Analysis (Interactive Graphic VS Static Graphic)		
	Static	Graphic
Decision Time	167	129
Decision Accuracy	89%	94%
Cognitive Load	3.04	2.18
Usefulness	4.02	4.18
Post Task Confidence	4.25	4.4
Visual Design	3.4	3.5
Emotion	3	2.7
Mean Difference Analysis (Interactive Tabular VS Static Tabular)		
	Static	Graphic
Decision Time	204	183
Decision Accuracy	77%	83%
Cognitive Load	3.43	2.84
Usefulness	3.81	4.2
Post Task Confidence	3.7	4.2
Visual Design	3.2	3.3
Emotion	2.7	2.5

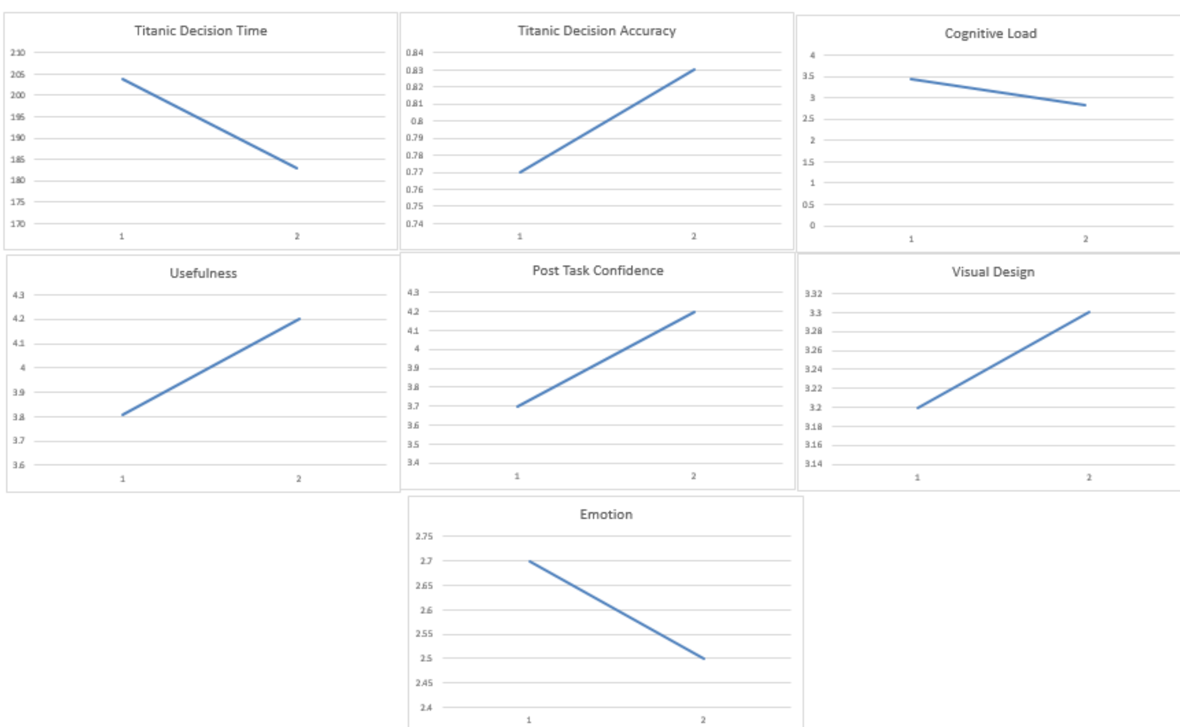


Figure 19. Respondent tabular static (Column 1) versus tabular interactive(Column 2) mean differences.

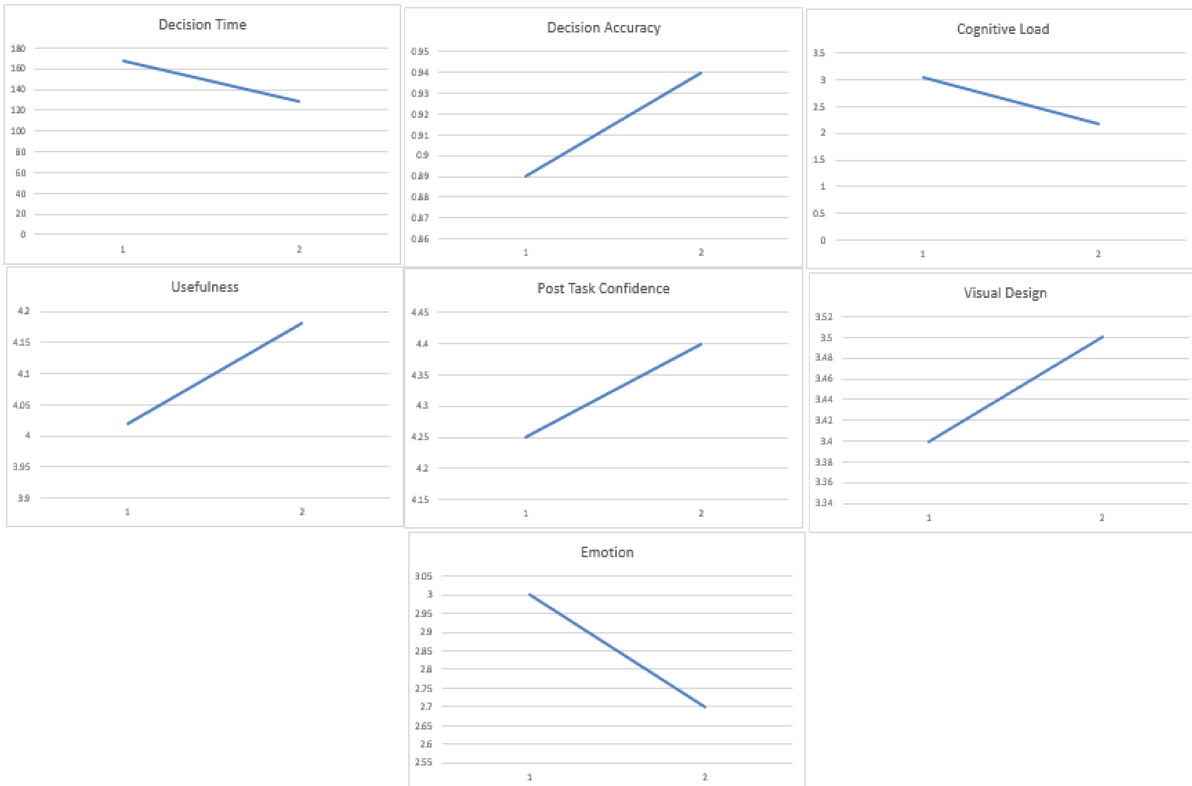


Figure 20. Respondent graphical static (Column 1) versus graphical interactive (Column 2) mean differences.

Table 21

Hypotheses Evaluation

Hypotheses		(CFA)
H1	Information presentation in a visual format will have a positive effect on perceived usability	Supported***
H2	Information presentation in a visual format will have a positive effect on task performance	Supported***
H3	Attitude towards task will have a positive effect on usability	Supported***
H4	Emotion elicited by the information presentation will have a positive effect on cognitive load	Supported***
H5	Post Task confidence will have a positive effect on task performance.	Supported***
H6	Cognitive load will have a negative effect on task performance.	Supported***
H7	Perceived usability of the information presentation will have a negative effect on cognitive load	Supported***
H8	Perceived visual design will have a positive effect on perceived usability	Supported***

H9	Perceived visual design will have a negative effect on emotion	Supported***
H10	Information presentation in a visual format will have a negative effect on cognitive load	Supported***
H11	Cognitive Load will have a negative effect on Post Task Confidence	Supported***
H12	Perceived usability will have a negative effect on post task confidence	Supported***

Conclusion

This research identifies and fills a TTF/DV gap in the IS literature by examining, in a contemporary end-user context, both the seminal TTF and tabular versus graphic image research. This examination results in the development of a new TTF/DV task performance model that extends TTF3. Our extended model (TTF/DV task performance model) combines utilization and fit with TTF3 to describe current concepts that enhance performance. The new model also embraces user attitudes and perceptions of the DV task and environment. We test the new model, measuring task performance outcomes within the bounds of a DV decision making process. Findings indicate that DV improves TTF, and allows individuals to respond faster and more accurately to the task, while demanding less cognitive load. This research contributes a new TTF/DV task performance model and provides evidence that, as a result of better task technology fit, DV improves decision effectiveness and task performance. These findings support the following contributions regarding the underlying seminal research.

First, this study contributes by affirming the seminal TTF model, as extended to include user attributes and perceptions of the contemporary DV task and its environment, is relevant to both TTF and DV research. This study indicates users perceive and process tabular and graphical information using different levels of cognitive effort (Lindgaard, Fernandes, Dudek, & Brown, 2006; Tufte & Graves-Morris, 1983). The findings suggest that decision makers expend more cognitive effort assimilating data depicted via tabular DV and less cognitive effort processing graphical DV (Daniel, 1992; Dickson, 2005; Feyyad, 1996; Robey & Farrow, 1982; Tufte & Graves-Morris, 1983). Therefore, this research suggests that a lower cognitive load enhances user attitudes and perceptions about the task and ultimately improving task performance. Additionally, our new extended TTF/DV task performance model suggests that graphical DV representations are more useful than tabular alternatives.

Second, this study contributes by affirming the seminal table versus graph research, as extended to include contemporary DV and associated technology, also remains relevant to researchers in the post-millennial era. Researchers conducted the early IS visual studies without a desktop OS that allowed user access to DV tools. Currently, visual tasks are more easily performed by users with the DV tools (Tableau and JMP) used in this study. The supported hypotheses verify the major premise of this study, that information presentation, attitudes, perceptions and graphical visual design of the DV task and environment improve overall fit to enhance task performance outcomes.

The results of this study show that the interactive DV no matter whether it was graphical or tabular, enhanced task performance as well as the core constructs of our model (usability, cognitive load, emotion, post task confidence). This is in line with theory that interactive visuals are more useful since they allow users to navigate the data themselves, and provide for discovery, especially when dealing with large amounts of data.

Summary Discussion

The goal of these studies is to address a task technology fit (TTF)/DV gap in the IS literature by jointly examining TTF and tabular versus graphic image research within the context of new technology. This examination allows developing and testing a new TTF/DV task performance model that extends TTF to include user attitudes and perceptions of the contemporary DV task. In order to accomplish this, the study integrates new literature to posit and test a TTF/DV model.

The results of these studies show that graphical DV whether Interactive or static is superior to equivalent tabular interactive or static DV in both accuracy and speed as well as the core constructs measured in this study (cognitive load, task confidence, emotion, and usability). Additionally, DV, as shown by these studies, can be used to improve fit and decision making performance, including tack attitudes, confidence, design aesthetics, and cognitive load. This research also highlights the importance of information presentation and individual difference, and how these individual differences enhance the decision making process.

This research also fills a gap by integrating DV concepts with TAM, TTF, cognitive fit, and individual differences, and by describing the benefits of graphical DV on performance. This study also introduces new knowledge into the literature by exploring a new theoretical model for DV decision making, new perspectives on fit and the role of individual differences, new perspectives on DV versus tabular decision making research and finally, this study offers new perspectives on tabular versus interactive research. Furthermore, this research extends decision research, by showing strong benefits from information presentation with design aesthetic stimuli, reducing cognitive load by using graphical DV, and improving decisions and performance with graphical DV

These findings indicate the dynamic and complex nature of the decision process, and highlight the multifaceted processes within decision making. This study extends the body of knowledge by creating and empirically testing a decision making success model based on the theories of technology acceptance model, task technology fit, and cognitive fit. Most prior research on the concept of decision making and decision support systems focuses heavily on one aspect or another of the decision making process, The aim of this study, which we feel was achieved is to take a broader look at the decision making process, to explore and bring attention to the many variables and relationships in existence in the process through the use of interactive DV.

This study highlights the influence of information presentation and individual differences in the decision making process. It also suggests that task attitude, confidence, design aesthetics, and cognitive load do indeed impact task performance as well. This research suggests that the use of graphical interactive DV as opposed to tabular interactive DV enhance task performance as well as usability, post task confidence, and cognitive load. Indicating the simply using any interactive DV is not sufficient in enhancing task performance. There are notable improvements with a graphical interactive DV aid.

This study is especially applicable to not only researchers but to practitioners as well. As decision support systems and business intelligence solutions become ubiquitous in industry, the importance of understanding the decision making process, and the many dimensions that contribute to success, cannot be emphasized enough. This model illuminates the “fit” between the decision maker and the task in an attempt to find a meaningful and successful partnership in order to facilitate better decision making.

Implications

This research finds that users both prefer and perform better with graphical DV over tabular DV, in addition to deriving a number of additional benefits from graphical DV. Thus, understanding how graphically designed DVs can amplify BI is indispensable to both IS researchers and practitioners.

For researchers, this study supports the need for continued IS research in DV, but it also provides compelling evidence for research in aesthetic visual design, which has received limited attention in the IS field. Building upon Zmud's (1979) work, adjusting the system for individual differences and task perceptions leads to a better task fit, which is achieved through graphical DV in our model. The graphical DV task fit affirmed in this study not only provides theoretical support for enhanced task outcomes, but tangibly yields greater accuracy, greater speed, and less cognitive load than for the tabular DV task. Looking forward, this work facilitates IS research to embrace the pre-cognitive world of visual design and incorporate it into the DV and BI decision making design process.

For practitioners, this study is especially applicable as DV and BI solutions become ubiquitous in industry (Gartner, 2013). The fundamental necessity of developers and managers understanding the decision making process, its many dimensions that contribute to a successful task fit, cannot be emphasized enough. The fit between the decision maker and the task creates favorable task outcomes, faster and more accurate performance, lower cognitive load, ,which contributes to less stress, (Lazarus & Folkman, 1984), and higher post task confidence. The user is more productive, more confident in their decisions, and more content with their work.

A broad implication of this study is the imminence of a successor paradigm, where millennials expect to intuitively understand and navigate applications without being technology experts. For it to be generation appropriate a system must defined by its interface; functionality and navigation recede from the forefront and visual design assumes a role as critical as functionality.

APPENDIX

DATA VISUALIZATIONS USED IN SURVEYS

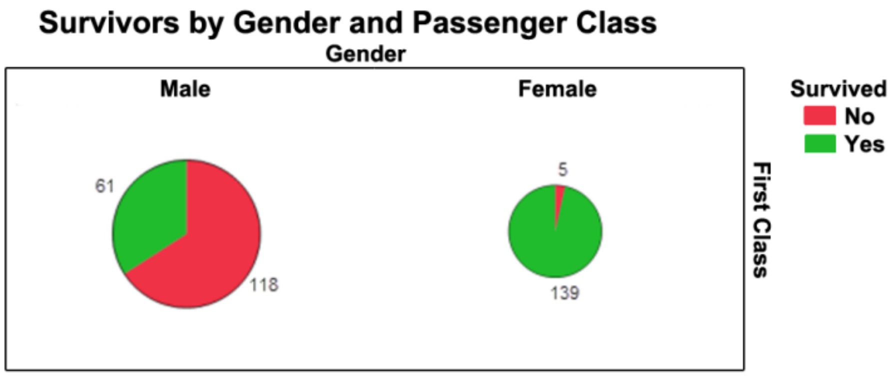
Graph Currently Showing: Passenger Class: All Gender: All Survived: All

Passenger Class	Didn't Survive	Survived
1st Class	123	200
2nd Class	158	119
3rd Class	528	181

Passenger Class
- (All)
- 1st Class
- 2nd Class
- 3rd Class

Survived?
- (All)
- Didn't Survive
- Survived

Passenger Sex
- (All)
- female
- male



Titanic Survivors by Gender and Passenger Class

Passenger Class 1	female	male
Survived(No)	5.0	118.0
Survived(Yes)	139.0	61.0

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